CG3201 Machine Learning & Deep Learning

Lecture 3 Bayesian Learning and Modelling

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Recap Lecture 2

- From linear classifiers to Support Vector Machines (SVM).
- Hard-Margin SVM: classify with margins.
  - Primal form.
  - Dual form from KKT conditions.
- Soft-Margin SVM: relaxing constraints for linearly inseparable data.
  - Primal form and dual form.
  - Converting constraints to losses via the hinge loss.
- Kernel SVM: separating data in the feature space with the kernel trick.
  - The kernel trick and SVM formulation with the kernel trick.
  - Kernel functions and impact of hyperparameters.
  - The Mercer’s theorem and the Mercer’s condition.

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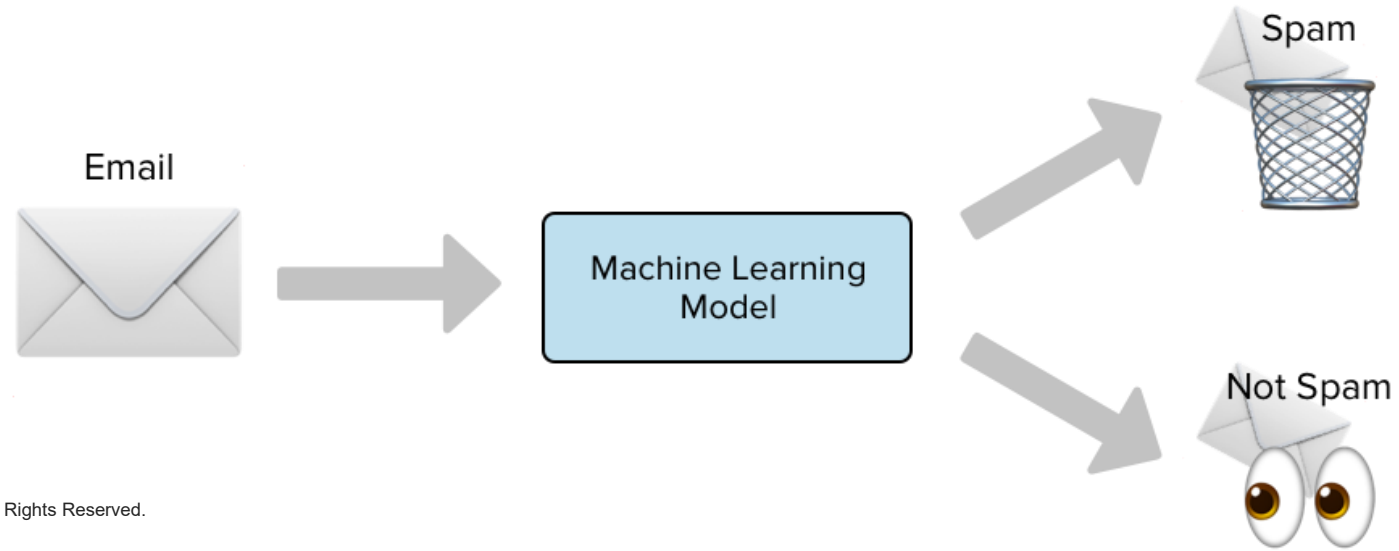
Comparing Models from Results

- In our discussion of SVM, we compare the different linear classifiers via the margins available for the classification:
  - The larger the margin, the better the SVM (assuming hard-margin SVM).
- In reality, we cannot really understand the exact features, not to say quantify the margins.
- Instead, we care more about the accuracy of the task (and sth else?)



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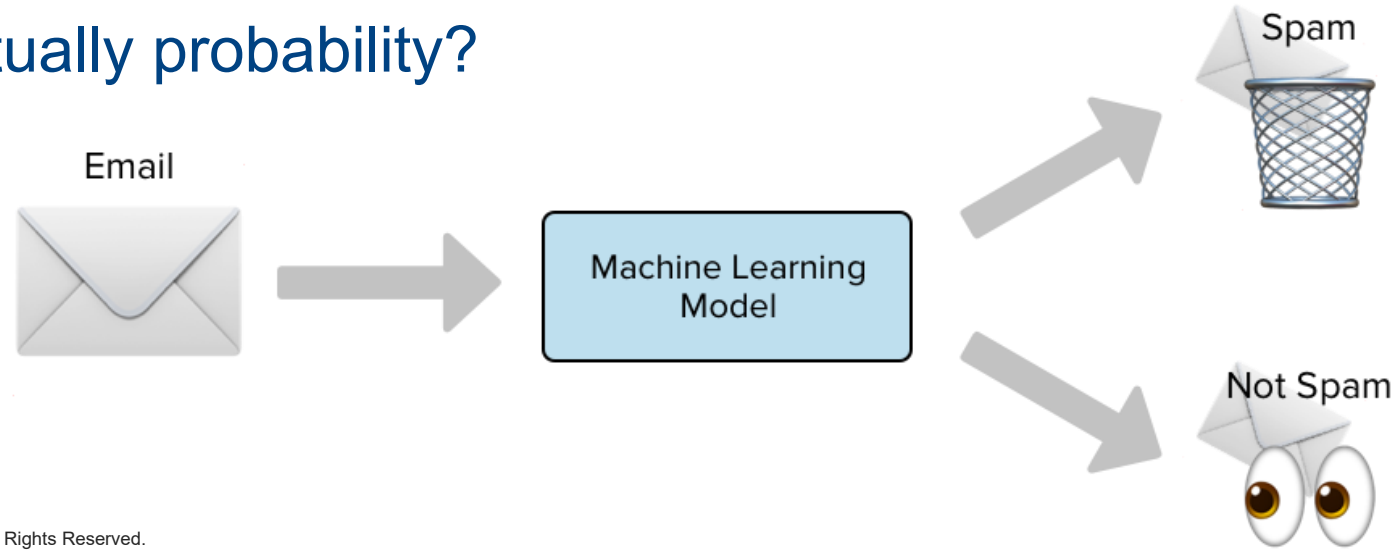
Comparing Models from Results

- Suppose we have a spam vs. ham email task.
  - Model A: of the 100 spams, it can correctly detect 70 of them.
  - Model B: of the 100 spams, it can correctly detect 80 of them.
  - Which one is better? What are we actually comparing?
- Probability. We say Model B has a higher probability of detecting spams when received.
- What is actually probability?



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Bayesian Learning Fundamentals 1:

- ✓ Introduction / Recall to Probability
- ✓ The Bayes' Rule
- ✓ Univariate Probability Distribution Models

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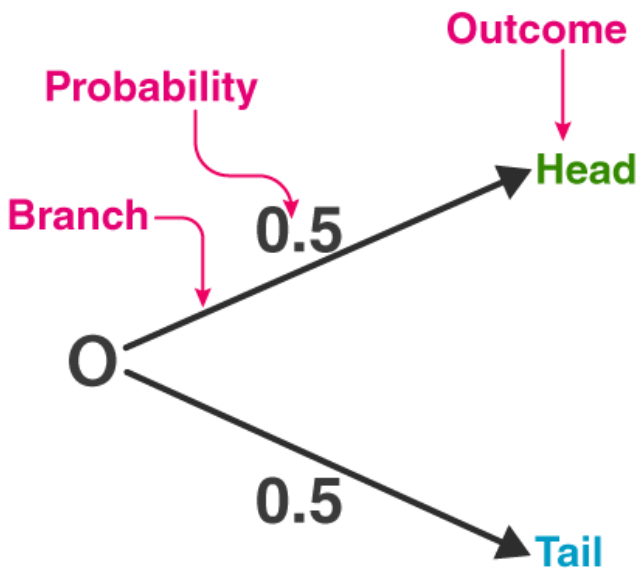
Probability: Two Interpretations

- Probability theory is nothing, but common sense reduced to calculation.
  – Pierre Laplace, 1812

- Frequentist interpretation: probabilities represent long run frequencies of events that can happen multiple times.
  - Flip the coin many times, we expect it to land heads about half the time.

- Bayesian interpretation: quantify **uncertainty** or ignorance about something
  - The coin is equally likely to land heads/tails on next toss.



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Properties of Probability of event(s): I

- Note: for this course, probability of an event denoted as: $P(A)$

- $P(A)$ satisfy: $0 \leq P(A) \leq 1$.
- Joint probability that events A and B both happening: $P(A, B)$;
- If A and B are independent events, $P(A, B) = P(A)P(B)$.
- Probability of event A or B happening (probability of union of A and B): $P(A \vee B) = P(A) + P(B) - P(A, B)$;
- If A and B are mutually exclusive (cannot happen at the same time), then $P(A \vee B) = P(A) + P(B)$;

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Properties of Probability of event(s): II

- Note: for this course, probability of an event denoted as: $P(A)$ or $p(A)$.

- Conditional probability of B happening given A has occurred:

$$P(B|A) \triangleq \frac{P(A,B)}{P(A)}$$

- Event A is independent of B if $P(A,B) = P(A)P(B)$;
- Events A ad B are conditionally independent given C if:

$$P(A,B|C) = P(A|C)P(B|C)$$

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Discrete and Continuous Random Variables (RV)

- A mathematical formalization of a quantity or object which depends on random events.
- A mapping or a function from possible outcomes in a sample space to a measurable space, often the real numbers.

Discrete variable	Continuous variable
Countable numbers of value (e.g., number of students)	Infinitely many real values (e.g., time to complete a task)
Distribution defined by probability mass function (pmf)	Distribution defined by cumulative distribution function (cdf) and probability density function (pdf)
$p(x) \triangleq P(X = x)$ $\sum p(x) = 1$	$P(x) \triangleq P(X \leq x)$
	$p(x) \triangleq \frac{d}{dx} P(x)$ $P(a < X \leq b) = \int_a^b p(x)dx = P(b) - P(a)$

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Discrete and Continuous Random Variables (RV)

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- Important: Note whether the variable is discrete or continuous! Could be confusing sometimes. Example:
  - A large quantity of green tea (e.g., 1000kg) is to be sold. X is the weight of green tea to be sold. Suppose any quantity can be sold, is X discrete or continuous?
  - What if the green tea can only be sold in packets of 10kg, is X now discrete or continuous?

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Bayes' Rule

- "Bayesian": refer to inference methods that represent "degrees of certainty" using probability theory, and which leverage Bayes' rule, to update the degree of certainty given data.
  - *Layman term: Update our belief based on prior belief and data so that we infer something with a certain degree of uncertainty*.
- Bayes' rule: a formula for computing the probability distribution over possible values of an unknown quantity H given observed data $Y = y$.

$$p(H = h|Y = y) = \frac{p(H = h)p(Y = y|H = h)}{p(Y = y)}$$

- Follows the product rule of probability: $p(h|y)p(y) = p(h)p(y|h) = p(h, y)$
- $p(H = h|Y = y)$ **: Posterior distribution / conditional distribution.**

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Bayes' Rule: in Detail

$$p(H = h|Y = y) = \frac{p(H = h)p(Y = y|H = h)}{p(Y = y)}$$

- $p(H)$: what we know about possible values of H before we see any data – the **prior distribution** (layman terms: prior belief).
- $p(Y|H = h)$: the distribution over possible outcomes Y we expect to see if $H = h$ – the **likelihood** (layman terms: the *observed* data).

- Multiplying the prior distribution and the likelihood – unnormalized joint distribution $p(H = h, Y = y)$. Convert into a normalized distribution by dividing by $p(Y = y)$, which is denoted as the **marginal likelihood**.
- $p(H = h|Y = y)$: represents our new **belief state** (updated belief) about the possible values of H – the **posterior distribution**.

$$\text{posterior} \propto \text{prior} \times \text{likelihood}$$

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Bayes' Rule: Example 1 – Weather Forecast

- Marie is getting married tomorrow at an outdoor ceremony in the desert.
- In recent years, it has rained only 5 days each year.
- When it actually rains, the weatherman has forecast rain 90% of the time.
- When it doesn't rain, he has forecast rain 10% of the time.
- Unfortunately, the weatherman is forecasting rain for tomorrow.
- What is the probability it will rain on the day of Marie's wedding?

$Y = 1\ (Y)$: *Weatherman forecast it rains*
$H = 1\ (H)$: *It Actually Rains*

Prior Probability

$$P(H) = \frac{5}{365} = 0.0137$$

Likelihood

$$P(Y|H) = 0.9$$

$$P(Y|\widetilde{H}) = 0.1$$

Posterior Probability

$$P(H|Y) = ??$$

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Bayes' Rule: Example 1 – Weather Forecast

- We know that: $P(H) = \frac{5}{365} = 0.0137$ and $P(Y|H) = 0.9$, $P\left(Y\middle|\widetilde{H}\right) = 0.1$.
  What we want: $P(H|Y)$.

$$P(H|Y) = \frac{P(H)P(Y|H)}{P(Y)} = \frac{P(H)P(Y|H)}{P(Y|H)P(H) + P(Y|\widetilde{H})P\left(\widetilde{H}\right)}$$

$$P(H|Y) = \frac{0.0137(0.9)}{0.9(0.0137) + 0.1(1 - 0.0137)} = 0.111$$

- The probability it will rain on the day of Marie's wedding, given the weatherman is forecasting rain for tomorrow is 0.111

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Bayes Rule: Example 2 – Disease Testing

- Suppose there is an outbreak of a disease (e.g., COVIDv2). Jack is taking a diagnostic test to determine if he is infected.
- Suppose the sensitivity (true positive rate) of the diagnostic is 87.5%, and the specificity (true negative rate) of the diagnostic is 97.5%.
- Also suppose the prevalence of the disease in the area where Jack lives is 10%.
- Unfortunately, Jack is tested positive. What is the probability of Jack being infected?

$Y = 1\ (Y)$: *Diagnostic test positive*
$H = 1\ (H)$: *Jack is infected*

Likelihood

$$P(Y|H) = 0.875$$
$$P(\tilde{Y}|\tilde{H}) = 0.975$$

Prior Probability

$$P(H) = 0.1$$

Posterior Probability

$$P(H|Y) = ??$$

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Bayes' Rule: Example 2 – Disease Testing

- We know that: $P(H) = 0.1$ and $P(Y|H) = 0.875$, $P\left(\tilde{Y}\middle|\tilde{H}\right) = 0.975$.
- What we want: $P(H|Y)$.

$$P(H|Y) = \frac{P(H)P(Y|H)}{P(Y)} = \frac{P(H)P(Y|H)}{P(Y|H)P(H) + P(Y|\tilde{H})P(\tilde{H})} = \frac{P(H)P(Y|H)}{P(Y|H)P(H) + \left(1 - P\left(\tilde{Y}\middle|\tilde{H}\right)\right)P(\tilde{H})}$$

$$P(H|Y) = \frac{P(H)P(Y|H)}{P(Y|H)P(H) + \left(1 - P\left(\tilde{Y}\middle|\tilde{H}\right)\right)\left(1 - P(H)\right)} = \frac{0.1 \times 0.875}{0.875 \times 0.1 + (1 - 0.1)(1 - 0.975)} = 0.795$$

- The chance of Jack being infected is 79.5%.

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Bayes Rule: Example 2 – Disease Testing

- Suppose there is an outbreak of a disease (e.g., COVIDv2). Jack is taking a diagnostic test to determine if he is infected.
- Suppose the sensitivity (true positive rate) of the diagnostic is 87.5%, and the specificity (true negative rate) of the diagnostic is 97.5%.
- Now suppose the prevalence of the disease in the area where Jack lives is changed to 1%.
- Jack is tested positive. What is the probability of Jack being infected now?

$Y = 1$ (Y): *Diagnostic test positive*
$H = 1$ (H): *Jack is infected*

Likelihood

$$P(Y|H) = 0.875$$
$$P(\tilde{Y}|\tilde{H}) = 0.975$$

Prior Probability

$$P(H) = 0.01$$

Posterior Probability

$$P(H|Y) = ??$$

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Bayes' Rule: Example 2 – Disease Testing

- We know that: $P(H) = 0.01$ and $P(Y|H) = 0.875$, $P\left(\tilde{Y}\middle|\tilde{H}\right) = 0.975$.
- What we want: $P(H|Y)$.

$$P(H|Y) = \frac{P(H)P(Y|H)}{P(Y)} = \frac{P(H)P(Y|H)}{P(Y|H)P(H) + P(Y|\tilde{H})P\left(\tilde{H}\right)} = \frac{P(H)P(Y|H)}{P(Y|H)P(H) + \left(1 - P\left(\tilde{Y}\middle|\tilde{H}\right)\right)P\left(\tilde{H}\right)}$$

$$P(H|Y) = \frac{P(H)P(Y|H)}{P(Y|H)P(H) + \left(1 - P\left(\tilde{Y}\middle|\tilde{H}\right)\right)\left(1 - P(H)\right)} = \frac{0.01 \times 0.875}{0.875 \times 0.01 + (1 - 0.01)(1 - 0.975)} = 0.261$$

- The chance of Jack being infected is 26.1%, even if tested positive.

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Bernoulli Distribution

- One of the simplest probability distribution – can be used to model binary events. (E.g., coin flipping)
  - Explaining the Bernoulli distribution with coin flipping. Denote $Y = 1$ as coin lands on heads, and $Y = 0$ as coin lands on tails.
  - Suppose the probability of the coin lands on heads is $0 \leq \theta \leq 1$. Thus, $p(Y = 1) = \theta$ and $p(Y = 0) = 1 - \theta$. Then random variable Y follow the Bernoulli distribution: $Y \sim \mathrm{Ber}(\theta)$.
  - $\mathrm{Ber}(y|\theta) = \begin{cases} 1 - \theta & if\ y = 0 \\ \theta & if\ y = 1 \end{cases} \triangleq \theta^y (1 - \theta)^{1-y}$

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Binomial Distribution

- Now, if we flip a coin multiple times. Then the sum of the Bernoulli random variables will follow a **Binomial** distribution.
  - The Bernoulli distribution is a **special case** of the binomial distribution.

  - Suppose we observe a set of N Bernoulli trials (e.g., toss a coin N times), $y_n \sim \mathrm{Ber}(\cdot|\theta)$ for $n = 1{:}N$. Define s to be the total number of heads $s \triangleq \sum_{n=1}^{N} \mathbb{I}(y_n = 1)$.
  - The distribution of s follows the binomial distribution:

$$\mathrm{Bin}(s|N,\theta) \triangleq \binom{N}{s} \theta^s (1-\theta)^{N-s} = \boldsymbol{C}_s^N \theta^s (1-\theta)^{N-s}$$

$$\text{binomial coefficient: } \binom{N}{s} \triangleq \frac{N!}{(N-s)!\,s!}$$

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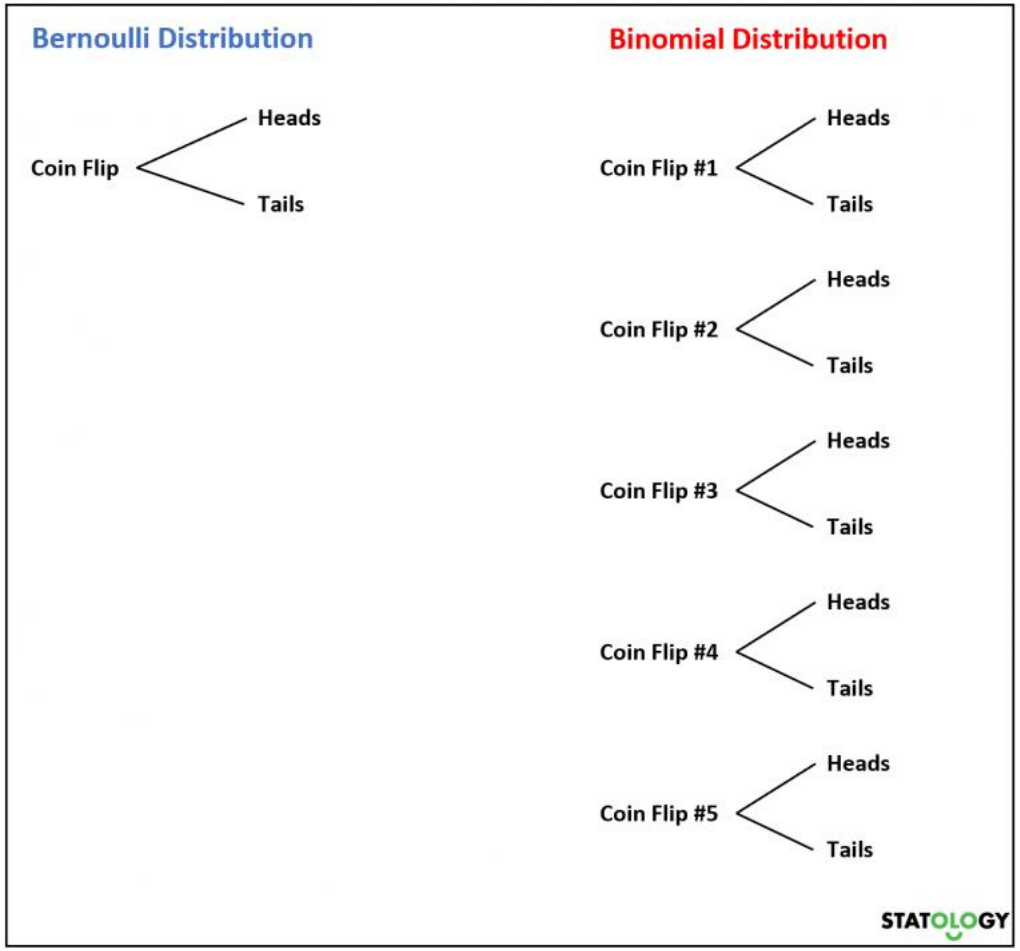
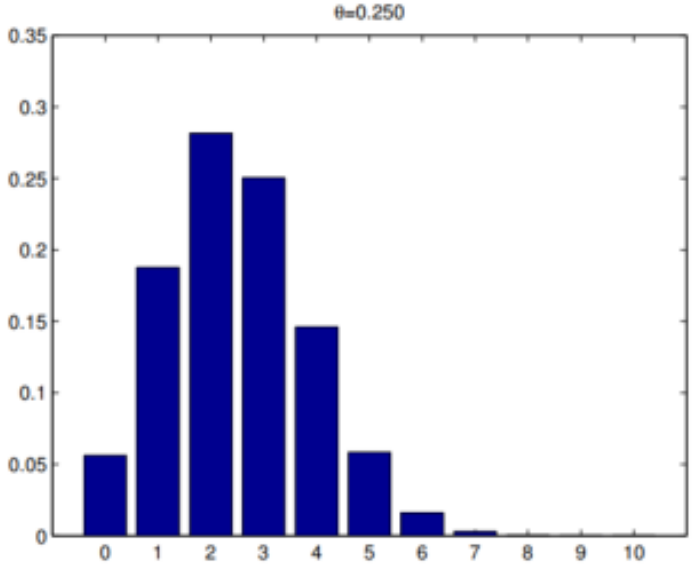
Binomial Distribution





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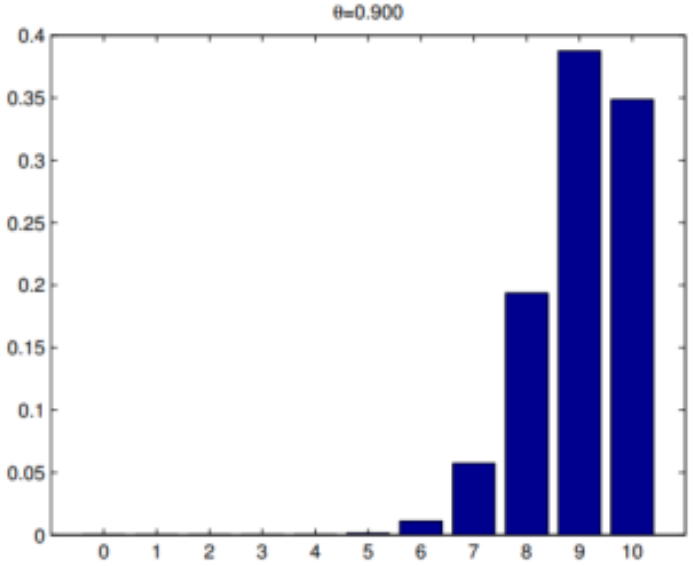

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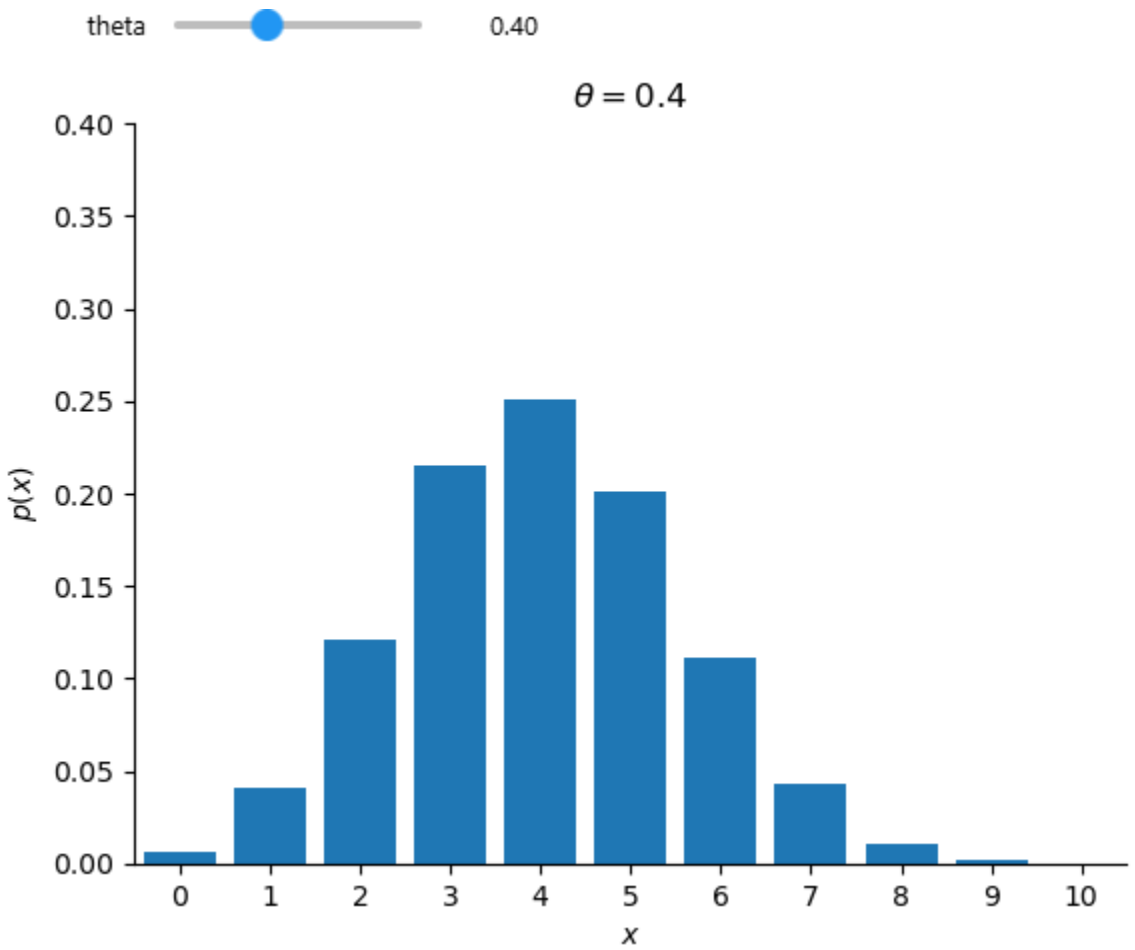
Binomial Distribution: Example



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Predicting a Binary Variable

- When we want to predict a binary variable $y \in \{0,1\}$ given input $x \in \mathcal{X}$, we need to use a conditional probability distribution:
  - $p(y|\boldsymbol{x}, \boldsymbol{\theta}) = \mathrm{Ber}(y|f(\boldsymbol{x}; \boldsymbol{\theta}))$.
  - $f(\boldsymbol{x}; \boldsymbol{\theta})$ is a function that predicts the parameter of the distribution.

- To avoid requiring $0 < f(\boldsymbol{x}; \boldsymbol{\theta}) \leq 1$, let f be an unconstrained function, we use the following model:
  - $p(y|\boldsymbol{x}, \boldsymbol{\theta}) = \mathrm{Ber}(y|\sigma(f(\boldsymbol{x}; \boldsymbol{\theta})))$.
  - $\sigma()$ is the sigmoid (logistic) function – "sigmoid" means S-shaped.

$$\sigma(a) \triangleq \frac{1}{1 + e^{-a}}$$

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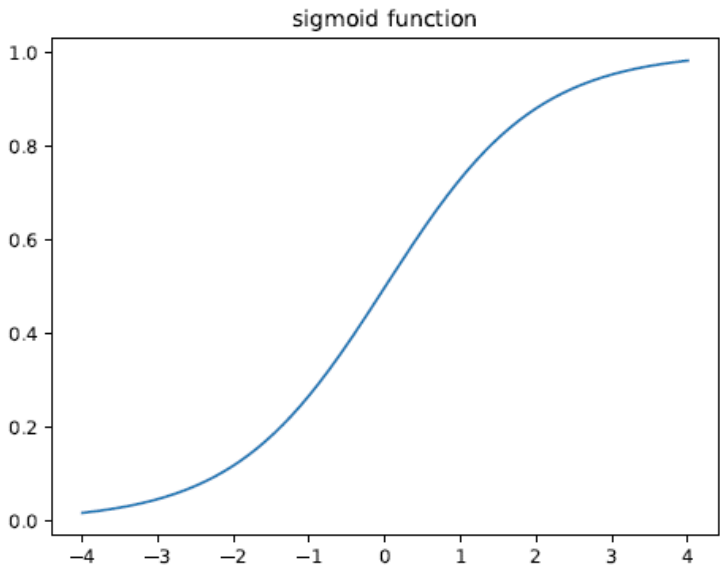
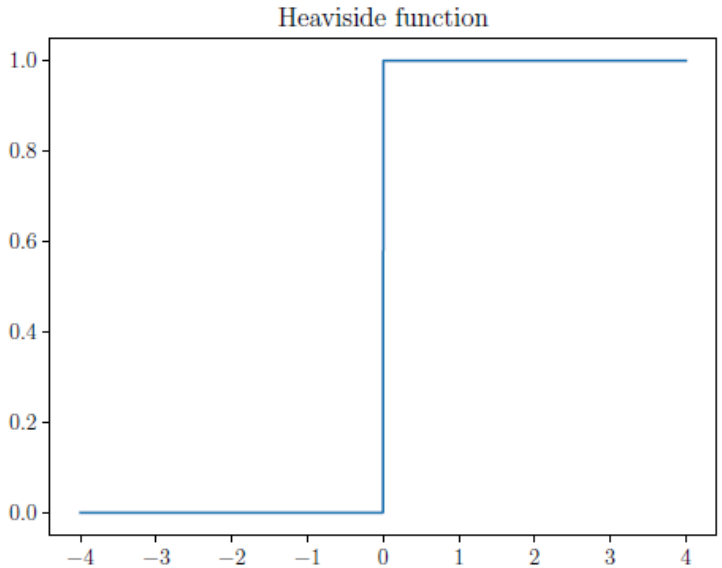
The Sigmoid Function Visualized

- Comparing the sigmoid function $\sigma(a) = (1 + e^{-a})^{-1}$ and the Heaviside step function $\mathbb{I}(a > 0)$.
- The sigmoid function can be thought of as a “soft” version of the heaviside step function.





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Categorical and Multinomial Distribution

- Problems for Bernoulli and Binomial distributions?
  - Can only model the probability for an event with ONLY 2 outcomes.
  - Generalize the Bernoulli to $C > 2$ values.
- **Categorical distribution**:
  - A discrete probability distribution with one parameter per class:
  - $\mathrm{Cat}(y|\boldsymbol{\theta}) \triangleq \prod_{c=1}^{C} \theta_c^{\mathbb{I}(y=c)}$, thus $p(y = c|\boldsymbol{\theta}) = \theta_c$.
  - Alternatively, convert the discrete variable y into a one-hot vector $\boldsymbol{y}$ with C elements.
  - E.g., if $C = 3$, then the classes are encoded as $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$.
  - As such, the categorical distribution is also written for $\boldsymbol{y}$:

$$\mathrm{Cat}(\boldsymbol{y}|\boldsymbol{\theta}) \triangleq \prod_{c=1}^{C} \theta_c^{y_c}$$

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Categorical and Multinomial Distribution

- The categorical distribution is a special case of **multinomial distribution**.
  - Repeat the categorical trials N times (e.g., roll a C-sided dice N times).
  - Define y_c to be the total number of times face c shows up: $y_c = N_c \triangleq \sum_{n=1}^{N} \mathbb{I}(y_n = c)$. Define $\boldsymbol{y}$ as the vector with C-dim, each element being y_c.
  - The distribution of $\boldsymbol{y}$ follows the multinominal distribution. Note that $\boldsymbol{y}$ is no longer a one-hot vector.

$$\mathcal{M}(y|N,\theta) \triangleq \binom{N}{y_1 \dots y_c} \prod_{c=1}^{C} \theta_c^{y_c} = \binom{N}{N_1 \dots N_c} \prod_{c=1}^{C} \theta_c^{N_c}$$

  - θ_c is the probability side c shows up and $\binom{N}{N_1 \dots N_c} \triangleq \frac{N!}{N_1! N_2! \dots N_C!}$ is the multinominal coefficient, which is the number of ways to divide a set of size $N = \sum_{c=1}^{C} N_c$ into subsets with sizes N_1 up to N_C. If $N = 1$, the multinomial distribution becomes the categorical distribution, if $C = 2$, the multinomial distribution is the binomial distribution.

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Predicting a Categorical Variable with Softmax

- When we want to predict a categorical variable y given input $x \in \mathcal{X}$, we need to use a conditional probability distribution:
  - $p(y|\boldsymbol{x}, \boldsymbol{\theta}) = \mathrm{Cat}(y|f(\boldsymbol{x}; \boldsymbol{\theta}))$.
- To avoid requiring $0 < f_c(\boldsymbol{x}; \boldsymbol{\theta}) \leq 1$ and $\sum_{c=1}^{C} f_c(\boldsymbol{x}; \boldsymbol{\theta}) = 1$, let f be an unconstrained function, we use the following model:
  - $p(y|\boldsymbol{x}, \boldsymbol{\theta}) = \mathrm{Cat}(y|\mathrm{softmax}(f(\boldsymbol{x}; \boldsymbol{\theta})))$.
  - $\mathrm{softmax}()$, also written as $\mathrm{sfm}()$ is the softmax (multinomial logistic) function

$$\mathrm{softmax}(\boldsymbol{a})_c \triangleq \left[\frac{e^{a_1}}{\sum_{c'=1}^{C} e^{a_{c'}}}, \dots, \frac{e^{a_C}}{\sum_{c'=1}^{C} e^{a_{c'}}}\right]$$

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Something More on Softmax

- 1. Properties of Softmax:
  - a. $\mathrm{softmax}(\boldsymbol{a})_c \in [0,1]$, and $\sum_{c=1}^{C} \mathrm{softmax}(\boldsymbol{a})_c = 1$. The softmax function has been applied to almost ALL classification tasks in deep learning thanks to these properties.
  - The input to the softmax $\boldsymbol{a} = f(\boldsymbol{x}; \boldsymbol{\theta})$ are called **logits**.
- 2. The softmax function, at its extremes, acts like the argmax function.
  - Divide each a_c (the components for each class of $\boldsymbol{a}$) by a constant T, denoted as the Temperature. Then as $T \to 0$, we find the following:
  - $\mathrm{softmax}(\boldsymbol{a}/T)_c = \begin{cases} 1.0 & \text{if } c = \arg\max_{c'} a_{c'} \\ 0.0 & \text{otherwise} \end{cases}$
  - At low temperatures, the distribution puts most of its probability mass in the most probable state (denoted as **winner takes all**), whereas at high temperatures, it spreads the mass uniformly.

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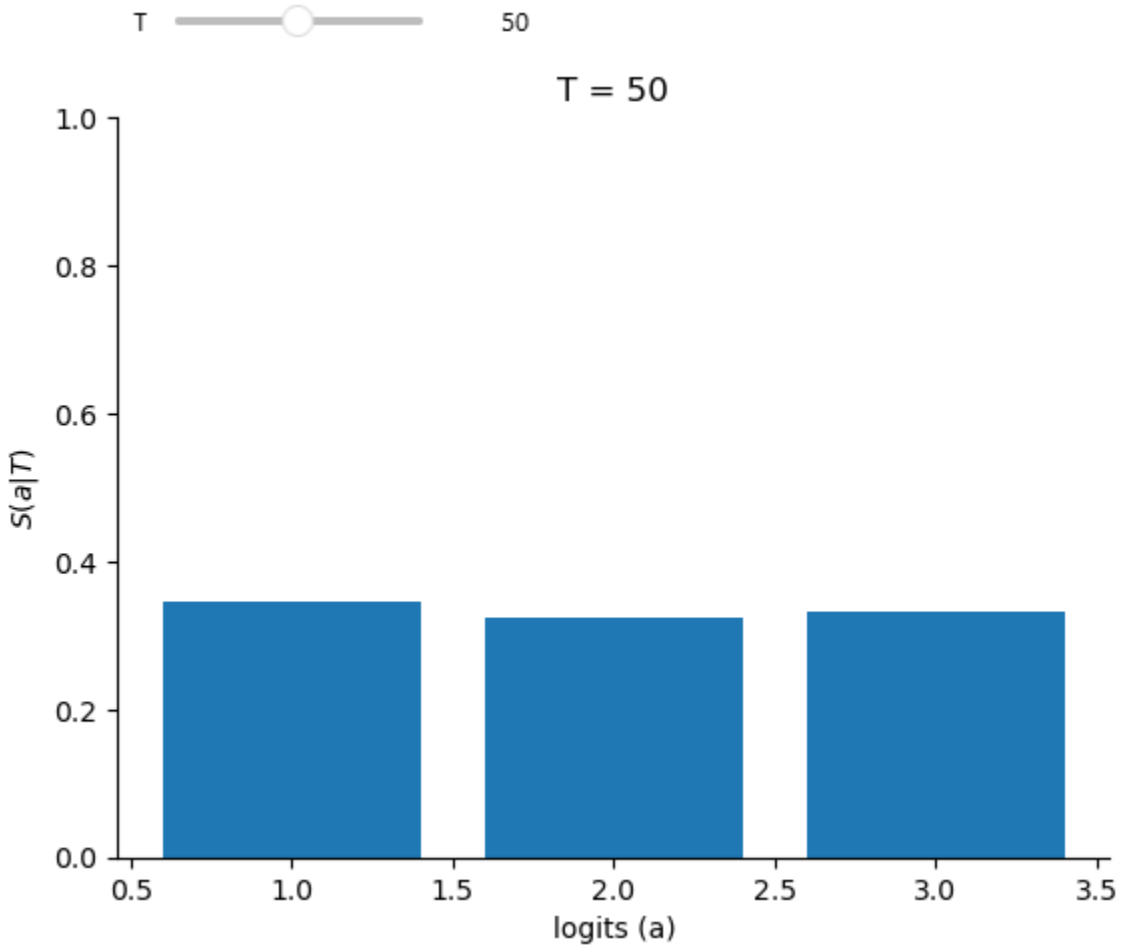
Illustration of Softmax Properties



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Univariate Gaussian Distribution

- The most widely used distribution of real-valued random variables $y \in \mathbb{R}$ is the Gaussian distribution (equivalently, normal distribution).
- Suppose a continuous variable X follow the Gaussian distribution, then cdf is: $P(x) \triangleq P(X \leq x)$ formulated as:

$$\Phi(x; \mu, \sigma^2) \triangleq \int_{-\infty}^{x} \mathcal{N}(z|\mu, \sigma^2) dz = \frac{1}{2}\left[1 + \mathrm{erf}\left(\frac{z}{\sqrt{2}}\right)\right]$$

- Here, μ encodes the mean of the distribution, σ^2 encodes the variance, when $\mu = 0$ and $\sigma = 1$, it refers to the **standard normal** distribution.
- $z = (x - \mu)/\sigma$ and $\mathrm{erf}(u)$ is the error function defined as:

$$\mathrm{erf}(u) \triangleq \frac{2}{\sqrt{\pi}} \int_{0}^{u} e^{-t^2} dt$$

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Univariate Gaussian Distribution

- The pdf is the derivative of the cdf: $p(x) \triangleq \frac{d}{dx} P(x)$ and the pdf of the Gaussian is given by:

$$\mathcal{N}(x|\mu, \sigma^2) \triangleq \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

- RV x that follows such pdf can be denoted as: $x \sim \mathcal{N}(\mu, \sigma^2)$.
- For Gaussian, the following results apply:
- Mean (expected value): $\mathbb{E}[\mathcal{N}(\cdot\,|\mu, \sigma^2)] = \mu$.
- Variance: $\mathbb{V}[X] = \mathbb{E}[X^2] - \mu^2$, therefore $\mathbb{E}[X^2] = \mu^2 + \sigma^2$.
- Standard deviation: $\mathrm{std}[X] = \sqrt{\mathbb{V}[X]} = \sigma$.

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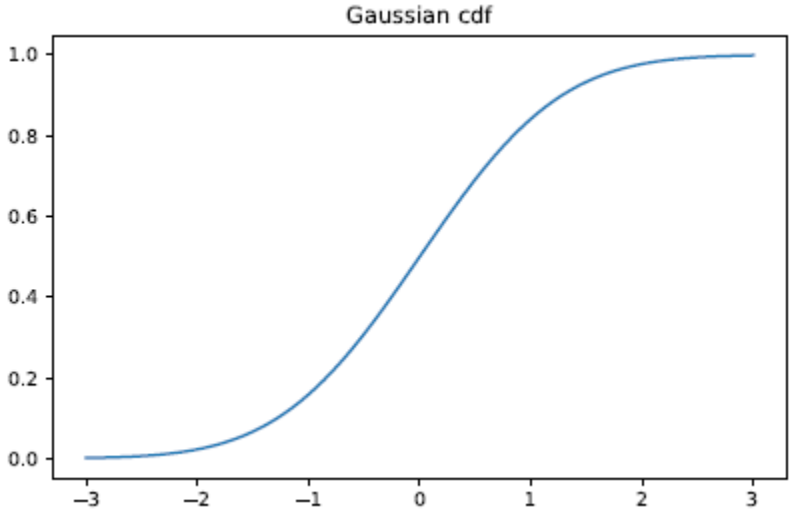
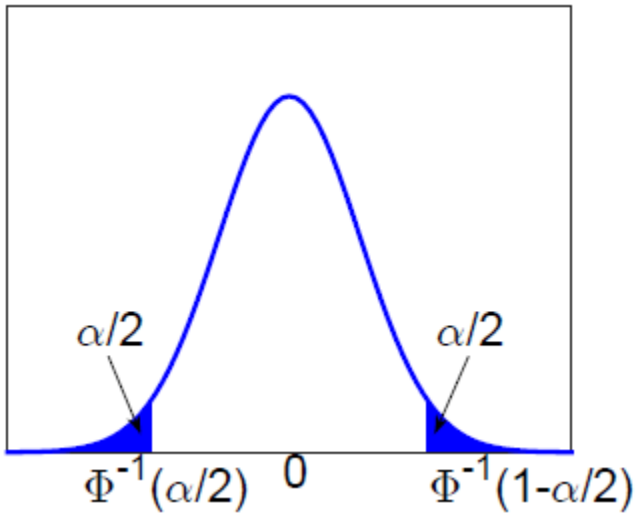
Univariate Gaussian Distribution

- The pdf is the derivative of the cdf: $p(x) \triangleq \frac{d}{dx} P(x)$ and the pdf of the Gaussian is given by:

$$\mathcal{N}(x|\mu, \sigma^2) \triangleq \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$





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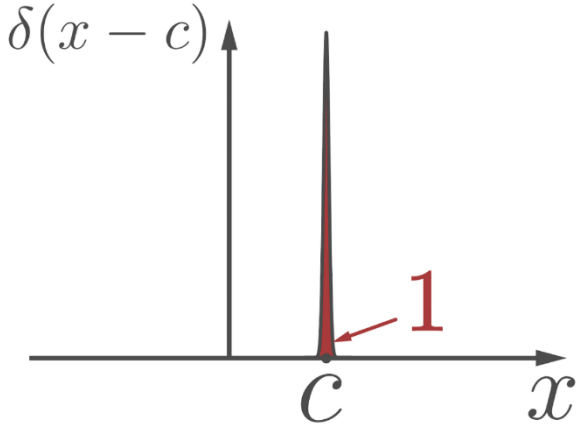
Other Common Univariate Distributions

- Dirac delta: when the σ of Gaussian → 0.

$$\lim_{\sigma \to 0} \mathcal{N}(y|\mu, \sigma^2) \to \delta(y-\mu) \quad \delta(x) = \begin{cases} +\infty & \text{if } x = 0 \\ 0 & \text{if } x \neq 0 \end{cases} \quad \int_{-\infty}^{\infty} \delta(x)dx = 1$$

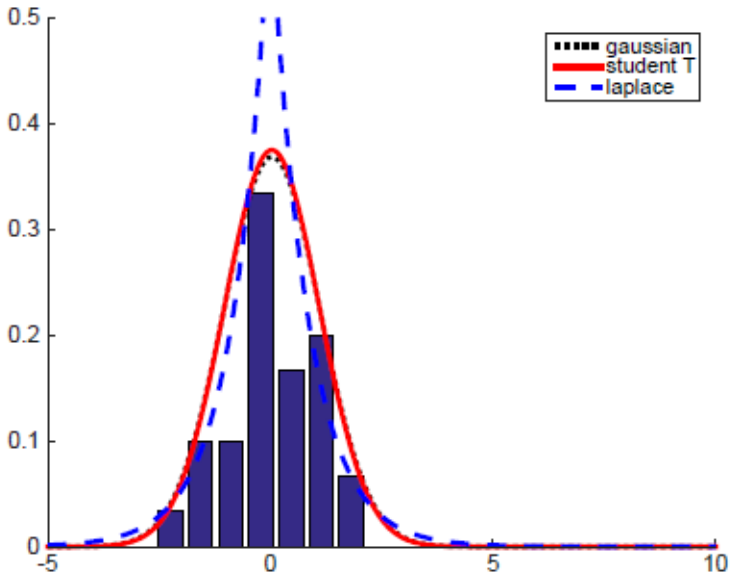


- Student t-distribution: less sensitive to outliers than the Gaussian distribution.

$$\mathcal{T}(y|\mu, \sigma^2, \nu) \propto \left[1 + \frac{1}{\nu}\left(\frac{y-\mu}{\sigma}\right)^2\right]^{-\left(\frac{\nu+1}{2}\right)}$$

- Laplace distribution: double-sided exponential distribution.

$$\text{Laplace}(y|\mu, b) \triangleq \frac{1}{2b}\exp\left(-\frac{|y-\mu|}{b}\right)$$



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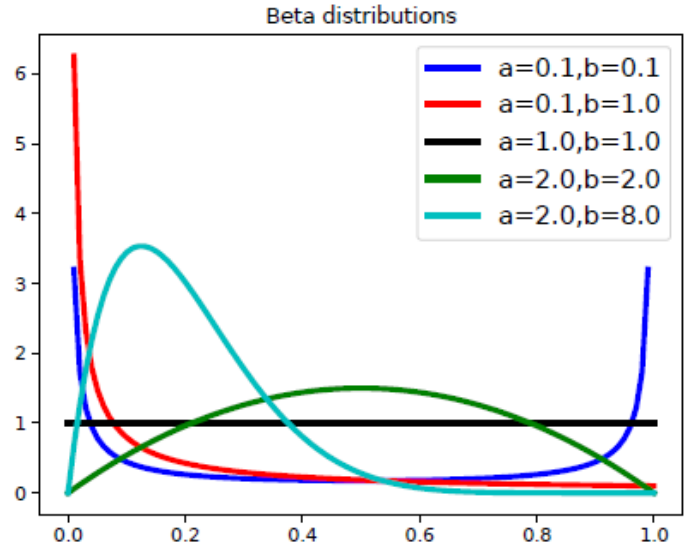
Other Common Univariate Distributions

- Beta distribution: a suitable model for the random behavior of percentages and proportions

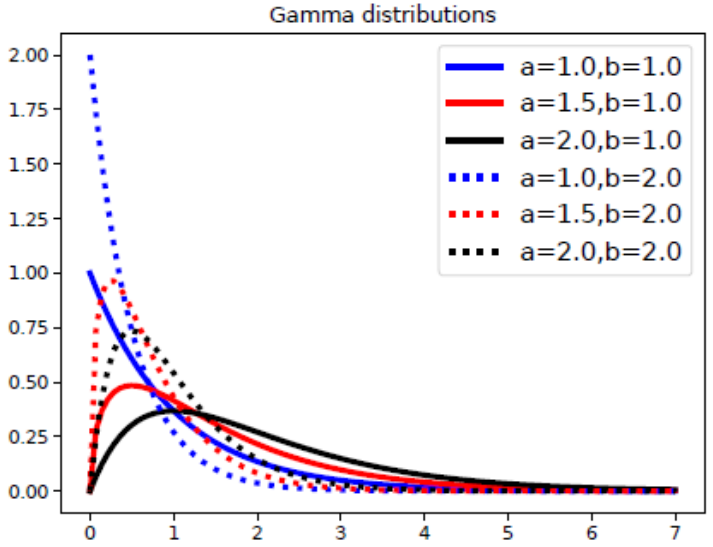
$$\mathrm{Beta}(x|a,b) = \frac{1}{B(a,b)} x^{a-1}(1-x)^{b-1}$$

$$B(a,b) \triangleq \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \quad \Gamma(a) \triangleq \int_0^\infty x^{a-1}e^{-x}dx$$



- Gamma distribution: a flexible distribution for positive real valued rv's.

$$\mathrm{Ga}(x|\mathrm{shape}=a, \mathrm{rate}=b) \triangleq \frac{b^a}{\Gamma(a)} x^{a-1}e^{-xb}$$



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Bayesian Learning and Modelling 1:

- ✓ Maximum Likelihood Estimation (MLE)
- ✓ MLE for Common Probability Distributions
- ✓ Empirical Risk Minimization

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Estimating Probability Models: Model Fitting

- Now suppose we know that the data would follow a certain probability model, controlled by parameters $\boldsymbol{\theta}$, but we don't know what the model exactly is.
  - Key problem: how to learn these parameters $\boldsymbol{\theta}$ from data?
  - The process of estimation θ from data $\mathcal{D}$ is called **model fitting**, or equivalently, "**training**" – the core of machine learning.
- The training boils down to an optimization of the form:

$$\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta})$$

  - $\widehat{\boldsymbol{\theta}}$: a point estimate; $\mathcal{L}(\boldsymbol{\theta})$: loss function / objective function.
  - Important: how to define the loss function / objective function?
  - Inference: quantify the uncertainty from data sample – compute $p(y|x, \widehat{\boldsymbol{\theta}})$.

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Maximum Likelihood Estimation (MLE)

- The intuitive objective: pick the parameters that assign the highest prob. to the training data (usually labelled) – maximum **likelihood** estimation:

$$\widehat{\boldsymbol{\theta}}_{mle} = \arg\max_{\boldsymbol{\theta}} p(\mathcal{D}|\boldsymbol{\theta})$$

  - Assume: training samples are independently sampled from the same distribution – iid assumption (independent and identical distributed).
  - $p(\mathcal{D}|\boldsymbol{\theta}) = \prod_{n=1}^{N} p(y_n|x_n, \boldsymbol{\theta})$ – likelihood; log likelihood takes its log $\ell(\boldsymbol{\theta}) \triangleq \log p(\mathcal{D}|\boldsymbol{\theta})$.
- Most optimization algorithms are designed to min cost functions (argmin instead of argmax), redefine the objective function to be the (conditional) negative log likelihood (NLL): maximize likelihood = minimize NLL.

$$\mathrm{NLL}(\boldsymbol{\theta}) \triangleq -\log p(\mathcal{D}|\boldsymbol{\theta}) = -\sum_{n=1}^{N} \log p(y_n|x_n, \boldsymbol{\theta})$$

$$\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}) = \arg\min_{\boldsymbol{\theta}} \mathrm{NLL}(\boldsymbol{\theta})$$

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MLE for Bernoulli Distribution

- Suppose Y is a random variable representing a coin toss, where the event $Y = 1$ corresponds to heads and $Y = 0$ corresponds to tails.
- Let $\theta = p(Y = 1)$ be the probability of heads. The probability distribution for this rv is the Bernoulli. The NLL for the Bernoulli distribution is given by:
  - Observe: this is just the empirical fraction of heads, an intuitive result!

$$
\begin{aligned}
\mathrm{NLL}(\theta) &= -\log \prod_{n=1}^{N} p(y_n|\theta) \\
&= -\log \prod_{n=1}^{N} \theta^{\mathbb{I}(y_n=1)}(1-\theta)^{\mathbb{I}(y_n=0)} \\
&= -\sum_{n=1}^{N} \left[\mathbb{I}\left(y_n = 1\right)\log\theta + \mathbb{I}\left(y_n = 0\right)\log(1-\theta)\right] \\
&= -[N_1 \log\theta + N_0 \log(1-\theta)]
\end{aligned}
$$

The MLE can be found by solving $\frac{d}{d\theta}\mathrm{NLL}(\theta) = 0$.

$$
\frac{d}{d\theta}\mathrm{NLL}(\theta) = \frac{-N_1}{\theta} + \frac{N_0}{1-\theta}
$$

and hence the MLE is given by

$$
\hat{\theta}_{\mathrm{mle}} = \frac{N_1}{N_0 + N_1}
$$

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MLE for Univariate Gaussian Distribution

- Suppose $Y \sim N(\mu, \sigma^2)$ and $\mathcal{D} = \{y_n : n = 1 : N\}$ is an iid sample of size N. The parameters are $\boldsymbol{\theta} = (\mu, \sigma^2)$. We estimate with MLE:

- 1. Derive NLL:

$$\mathrm{NLL}(\mu, \sigma^2)$$

$$= -\sum_{n=1}^{N} \log\left[\left(\frac{1}{2\pi\sigma^2}\right)^{\frac{1}{2}} \exp\left(-\frac{1}{2\sigma^2}(y_n - \mu)^2\right)\right]$$

$$= \frac{1}{2\sigma^2}\sum_{n=1}^{N}(y_n - \mu)^2 + \frac{N}{2}\log(2\pi\sigma^2)$$

- 2. Derive optimal parameter:

$$\frac{\partial}{\partial \mu}\mathrm{NLL}(\mu, \sigma^2) = 0, \quad \frac{\partial}{\partial \sigma^2}\mathrm{NLL}(\mu, \sigma^2) = 0$$

$$\hat{\mu}_{\mathrm{mle}} = \frac{1}{N}\sum_{n=1}^{N} y_n = \overline{y}$$

$$s^2 \triangleq \frac{1}{N}\sum_{n=1}^{N} y_n^2$$

$$\hat{\sigma}^2_{\mathrm{mle}} = \frac{1}{N}\sum_{n=1}^{N}(y_n - \hat{\mu}_{\mathrm{mle}})^2$$

$$= \frac{1}{N}\sum_{n=1}^{N}\left[y_n^2 + \hat{\mu}^2_{\mathrm{mle}} - 2y_n\hat{\mu}_{\mathrm{mle}}\right] = s^2 - \overline{y}^2$$

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Empirical Risk Minimization (ERM)

- The MLE is based on: $\mathrm{NLL}(\boldsymbol{\theta}) = -\sum_{n=1}^{N} \log p(y_n | x_n, \boldsymbol{\theta})$
  - Can we further generalize the MLE?
  - Recall: get optimal $\boldsymbol{\theta}$ by $\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}) = \arg\min_{\boldsymbol{\theta}} \sum_{n=1}^{N} \ell(\boldsymbol{\theta})$.
- Generalize MLE: replace log loss term with other loss function!
- $\ell(\boldsymbol{\theta}) = -\log p(y_n | x_n, \boldsymbol{\theta}) \rightarrow \ell(y_n, \boldsymbol{\theta}; x_n)$.
- With the more generalized form, we have:
- $\mathcal{L}(\boldsymbol{\theta}) = \sum_{n=1}^{N} \ell(y_n, \boldsymbol{\theta}; x_n)$ – Empirical risk minimization (ERM).
  - The expected loss where the expectation is taken w.r.t. the “empirical” (sampled) distribution.

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ERM: Minimizing Misclassification Rate

- If we are solving a classification problem, an intuitive loss selection would be the 0-1 loss: (f being some kind of predictor)

$$\ell_{01}(y_n, \boldsymbol{\theta}; x_n) = \begin{cases} 0 & if\ y_n = f(x_n; \boldsymbol{\theta}) \\ 1 & if\ y_n \neq f(x_n; \boldsymbol{\theta}) \end{cases}$$

- Then the empirical risk becomes:

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_{n=1}^{N} \ell_{01}(y_n, \boldsymbol{\theta}; x_n)$$

- This is defined as the empirical **misclassification rate** on the training set.

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Bayesian Learning and Modelling 2:

- ✓ Bayesian Decision Theory and the Classification Problem
- ✓ Maximum A Posteriori (MAP).
- ✓ Evaluation of Classifiers

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Decision Theory: Making the Right Decision

- More often, we don't actually know the type of distribution for the data either – not to mention the parameter of the distribution.
- How can we make decision then without such prior knowledge?

- Decision Theory: a set of quantitative methods for reaching optimal decisions.
  - Concerned with the reasoning underlying an agent's choices, whether this is a mundane choice between taking the bus or getting a taxi, or a more far-reaching choice about whether to pursue a demanding political career.
- Layman term: *how we can decide what action should be taken that could end up with the best result*?

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(Bayesian) Decision Theory: in Detail

- In decision theory, we assume the decision maker, or **agent**, has a set of possible **actions**, $\mathcal{A}$, to choose from.
- Example: doctor treat someone who may have cancer, actions $a \in \mathcal{A}$:
  - Do nothing;
  - Give patient expensive drugs with bad side effect but can save their life;
- We can compute (implicitly) the cost of treating a person (test $= 0$: cancer negative; age $= 0$: young age; action $= 1$: apply the drugs).

test	age	pr(cancer)	cost-noop	cost-drugs	action
0	0	0.01	0.84	8.00	0
0	1	0.01	0.14	8.00	0
1	0	0.80	47.73	8.00	1
1	1	0.80	7.95	8.00	0

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(Bayesian) Decision Theory: in Detail

test	age	pr(cancer)	cost-noop	cost-drugs	action
0	0	0.01	0.84	8.00	0
0	1	0.01	0.14	8.00	0
1	0	0.80	47.73	8.00	1
1	1	0.80	7.95	8.00	0

- Every action $a \in \mathcal{A}$ has cost and benefits, depending on the underlying state of nature $h \in \mathcal{H}$. How can we explicitly model such cost / benefit?
- Define: loss function $\ell(h, a)$ – loss incurred if action a is taken at state h.
  - The next question: how to leverage this loss?
  - Equivalently: how can we actually know quantitatively what is the best action if we compute the loss function?

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(Bayesian) Decision Theory: in Detail

- Define: loss function $\ell(h, a)$ – action a taken at state h.

- Compute a posterior expected loss (risk) for each possible a: (where the data is defined as x)

$$\rho(a|x) \triangleq \mathbb{E}_{p(h|x)}[\ell(h, a)] = \sum_{h \in \mathcal{H}} \ell(h, a) p(h|x)$$

- We thus define the **optimal policy** $\pi^*(x)$, which is also called the **Bayes estimator** or **Bayes decision rule** $\delta^*(x)$:

$$\pi^*(x) = \arg\min_{a \in \mathcal{A}} \rho(a|x) = \arg\min_{a \in \mathcal{A}} \mathbb{E}_{p(h|x)}[\ell(h, a)]$$

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Decision Theory: in Detail

- The **optimal policy** $\pi^*(x)$, or **Bayes decision rule** $\delta^*(x)$:

$$\pi^*(x) = \arg\min_{a \in \mathcal{A}} \rho(a|x) = \arg\min_{a \in \mathcal{A}} \mathbb{E}_{p(h|x)}[\ell(h, a)]$$

- If we define a **utility function** $U(h, a)$ as the desirability of each action at each state, then: $U(h, a) = -\ell(h, a)$
- The optimal policy can be re-written as:

$$\pi^*(x) = \arg\max_{a \in \mathcal{A}} \mathbb{E}_h[U(h, a)]$$

- This is defined as the **maximum expected utility principle**.
- Next, we use Bayesian decision theory to decide the optimal class label to predict given an observed input $x \in \mathcal{X}$.

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Classification with Zero-One Loss

- Set the following assumptions:
  - The states of nature correspond to class labels: $\mathcal{H} = \mathcal{Y} = \{1, \dots, C\}$;
  - The actions correspond to class labels: $\mathcal{A} = \mathcal{Y}$.
- Under the above assumptions, a common loss function (and a very simple one) is the **zero-one** loss $\ell(h, a) = \ell_{01}(y^*, \hat{y})$:

	$\hat{y} = 0$	$\hat{y} = 1$
$y^* = 0$	0	1
$y^* = 1$	1	0

- Or mathematically $\ell_{01}(y^*, \hat{y}) = \mathbb{I}(y^* = \hat{y})$

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Classification with Zero-One Loss

- The posterior expected loss $\rho(a|x) \triangleq \mathbb{E}_{p(h|x)}[\ell(h,a)]$:

$$\rho(a|x) = \rho(\hat{y}|x) = p(\hat{y} \neq y^*|x) = 1 - p(\hat{y} = y^*|x)$$

- The action that minimizes the expected loss is to choose the most probable action:

$$\pi(x) = \arg\max_{\hat{y} \in \mathcal{Y}} p(\hat{y}|x)$$

- This corresponds to the **mode** of the posterior distribution, also known as the **maximum a posteriori** or **MAP** estimate.

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Cost-Sensitive Classification

- Problem of the zero-one loss?
  - Cost of false positive (ℓ_{01}) = cost of false negative (ℓ_{10})
  - For cost-sensitive classification, the loss function would be:

$$\begin{pmatrix} \ell_{00} & \ell_{01} \\ \ell_{10} & \ell_{11} \end{pmatrix}$$

		Estimate		Row sum
		0	1	
Truth	0	TN	FP	N
	1	FN	TP	P
Col. sum		$\hat{N}$	$\hat{P}$	

  - Set $p_0 = p(y^* = 0|x)$ and $p_1 = 1 - p_0$
  - We would choose label $\hat{y} = 0$ iff $\ell_{00}p_0 + \ell_{10}p_1 < \ell_{11}p_1 + \ell_{01}p_0$.
  - Usually, $\ell_{00} = \ell_{11} = 0$. Thus, the above simplifies to: $p_1 < \frac{\ell_{01}}{\ell_{01}+\ell_{10}}$
  - Example (Try out): if a false negative costs twice as much as false positive, then we use a decision threshold of $1/3$ before declaring a positive.

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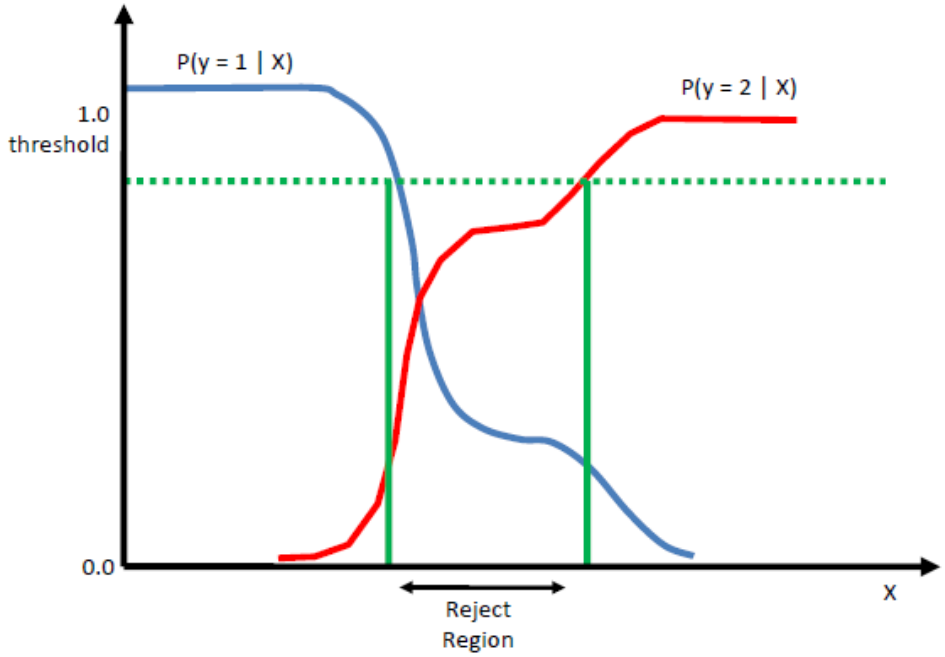
Classification with Reject Option

- We may be able to say "**I don't know**" instead of returning an answer that we don't really trust – choosing the **reject** option.
  - Important in domains such as medicine where we may be risk adverse.



https://www.youtube.com/watch?v=P18EdAKuC1U

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Classification with Reject Option

- Set the following assumptions:
  - The states of nature correspond to class labels: $\mathcal{H} = \mathcal{Y} = \{1, \dots, C\}$;
  - The actions correspond to class labels: $\mathcal{A} = \mathcal{Y} \cup \{0\}$.
  - The loss function is defined as:

$$l(y^*, a) = \begin{cases} 0, & \text{if } y^* = a \text{ and } a \in \{1, \dots, C\} \\ \lambda_r, & \text{if } a = 0 \\ \lambda_e, & \text{otherwise} \end{cases}$$

  - Here λ_r: cost of the reject action, and λ_e: cost of a classification error. Set $\hat{\lambda} = 1 - \lambda_r / \lambda_e$, we thus compute the optimal policy as:

$$a^* = \begin{cases} \hat{y}, & \text{if } p^* > \hat{\lambda} \\ \text{reject}, & \text{otherwise} \end{cases}$$

  - Here $\hat{y} = \arg\max_y p(y|x)$ and $p^* = p(y^*|x) = \max_y p(y|x) = p(\hat{y}|x)$.

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Class Confusion Matrices

- We can choose the optimal label in a binary classification problem by **thresholding** the probability – hard threshold!
  - Instead of picking a **single threshold**, consider using a set of different thresholds, and comparing the resulting performance.
- For a threshold τ, the decision rule: $\hat{y}_\tau(x) = \mathbb{I}(p(y = 1|x) > 1 - \tau)$.
  - $FP_\tau = \sum_{n=1}^{N} \mathbb{I}(\hat{y}_n(x_n) = 1, y_n = 0)$, similar for FN_τ, TP_τ, and TN_τ.

		Predicted condition	
	Total population = P + N	Positive (PP)	Negative (PN)
Actual condition	Positive (P)	True positive (TP), hit	False negative (FN), type II error, miss, underestimation
	Negative (N)	False positive (FP), type I error, false alarm, overestimation	True negative (TN), correct rejection

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- The simple form of evaluation: accuracy

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + FN + FP + TN)}$$

- Is this reliable?
  - If have imbalanced testing data (class 0 = 90, class 1 = 10), if the model predict **everything** as class 0, Accuracy = 90%

Class Confusion Matrices

- True positive rate (TPR), or sensitivity, recall, hit rate:

$$TPR_\tau = p(\hat{y} = 1 | y = 1, \tau) = \frac{TP_\tau}{TP_\tau + FN_\tau}$$

- False positive rate (FPR), or false alarm rate, type I error rate:

$$FPR_\tau = p(\hat{y} = 1 | y = 1, \tau) = \frac{FP_\tau}{FP_\tau + TN_\tau}$$

- Plot the TPR vs FPR as an implicit function of τ: Receiver Operating Characteristic (ROC) curve – Plot TPR vs FPR with varying τ.

		Predicted condition	
	Total population = P + N	Positive (PP)	Negative (PN)
Actual condition	Positive (P)	True positive (TP), hit	False negative (FN), type II error, miss, underestimation
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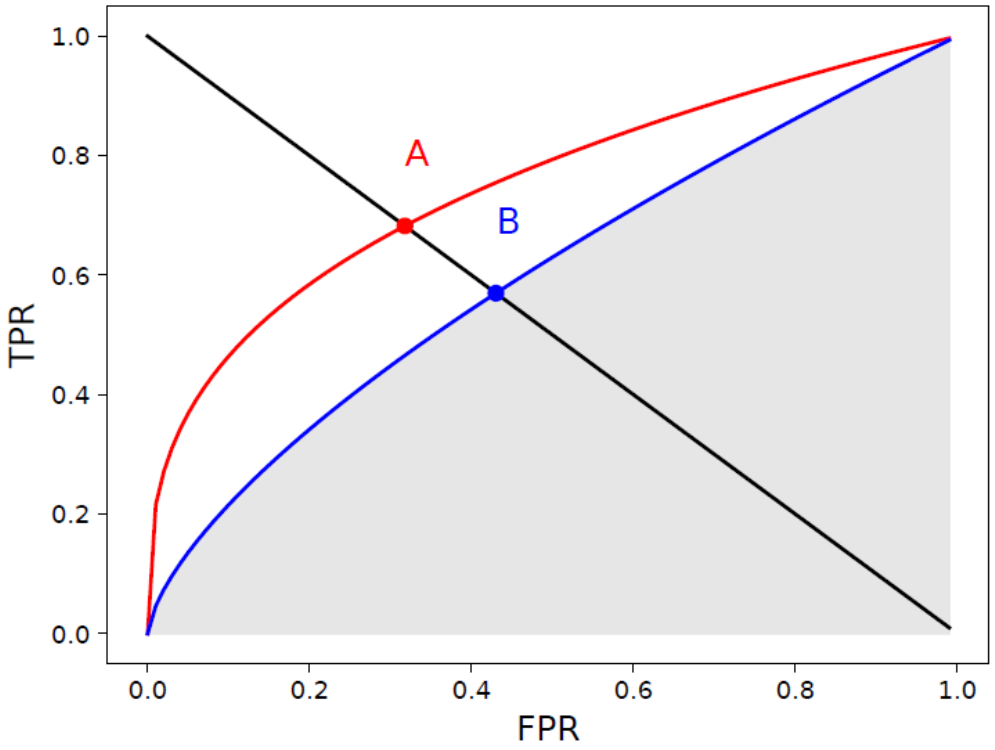

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Class Confusion Matrices

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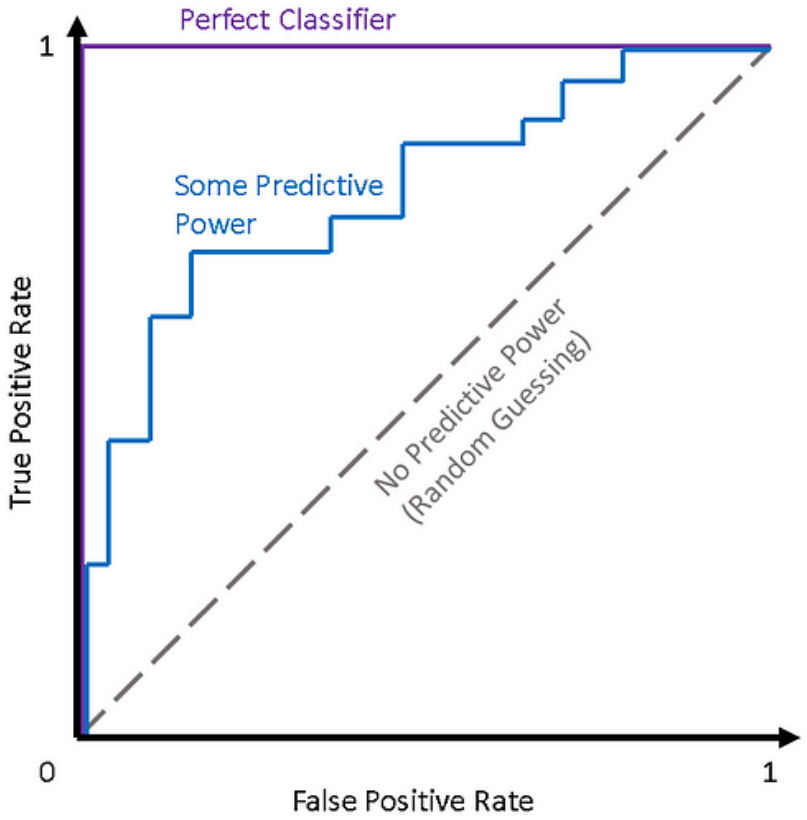

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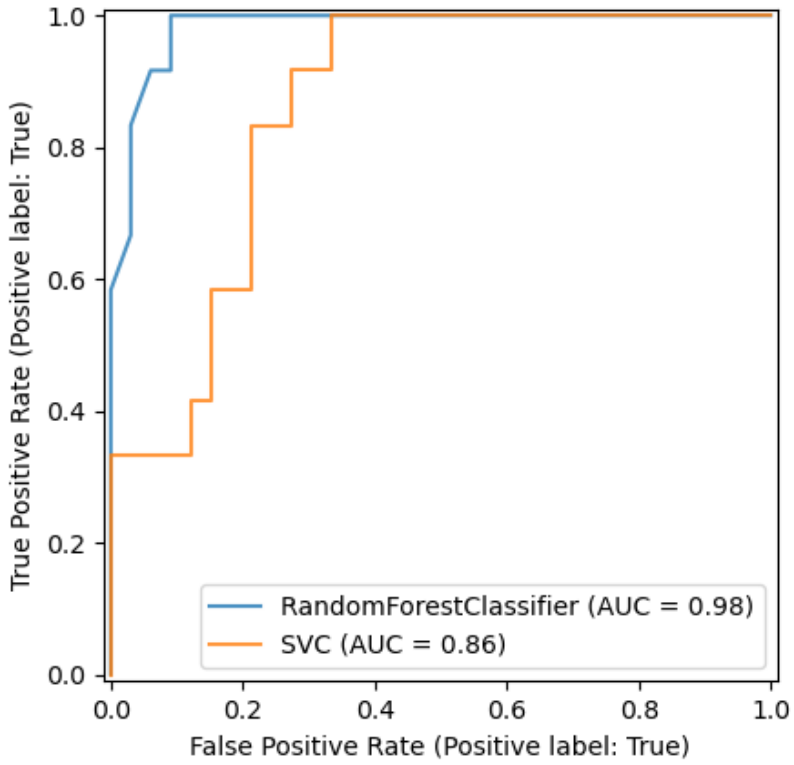
Plotting the ROC Curve: Example

- ROC curve – Plot the TPR vs FPR as an implicit function of τ, i.e., plot TPR vs FPR with varying τ.
- How do we compare different ROC curves?



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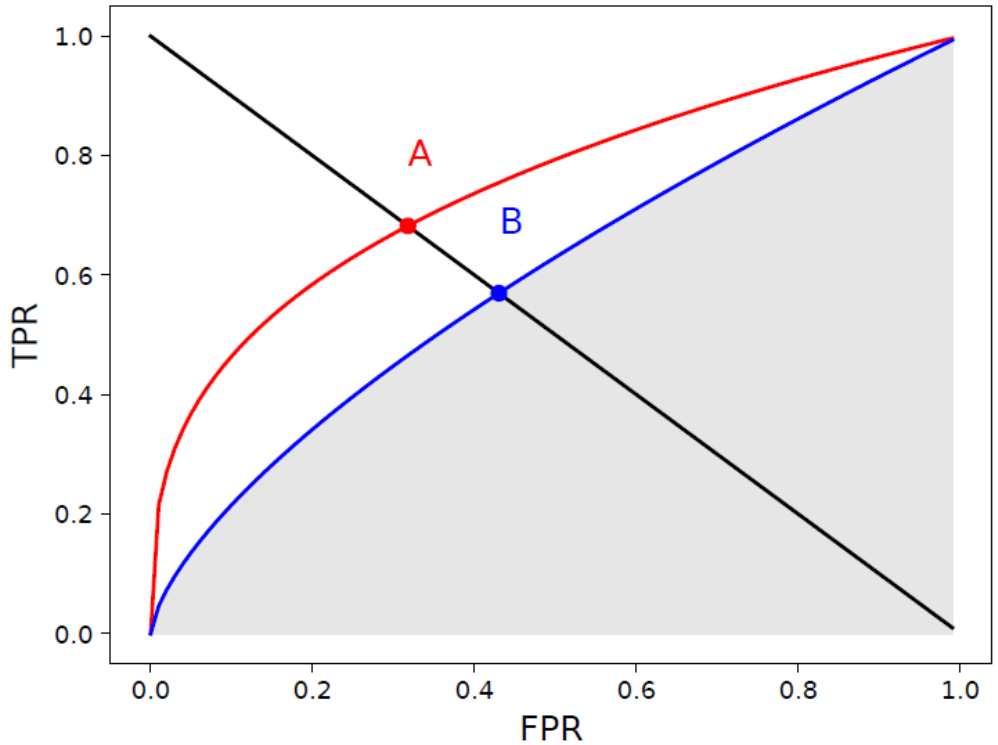
The ROC Curve

- Quality of a ROC curve is often summarized using the Area Under the Curve (AUC). Higher AUC scores better; max is 1.

- Another statistic is the Equal Error Rate (EER), or the cross-over rate
  - Defined as the value which satisfies $FPR = FNR = 1 - TPR$.
  - Compute the EER by drawing a line from the top left to the bottom right and seeing where it intersects the ROC curve.
  - Lower EER scores are better; the minimum is obviously 0 (top left corner).



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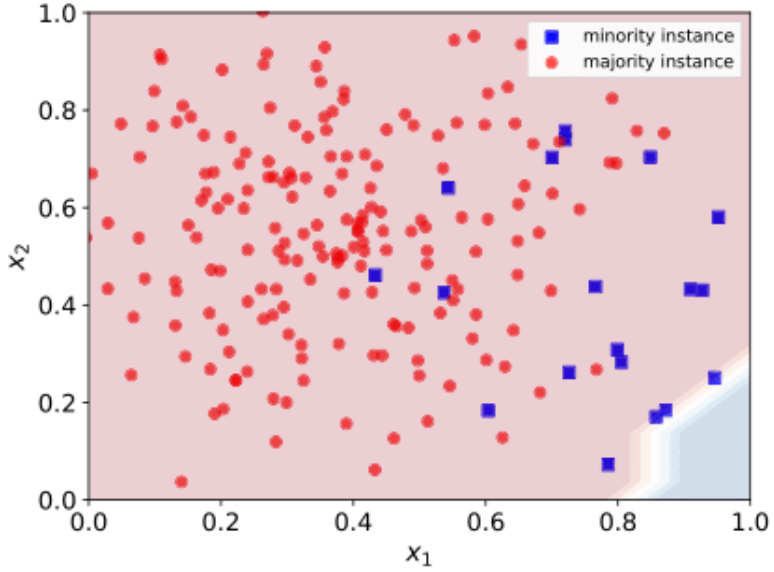
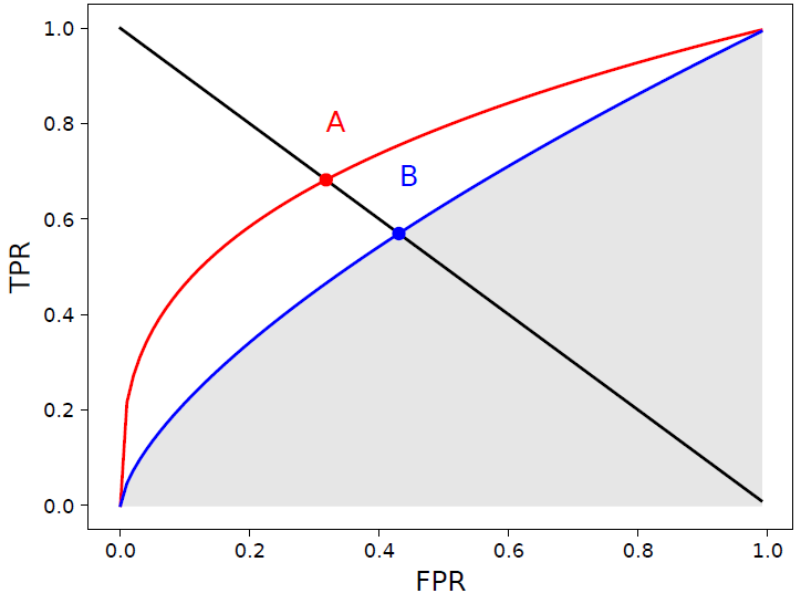
Class Imbalance and the ROC Curve

- In some problems, there is severe **class imbalance**.
  - E.g., in information retrieval, the set of negatives (irrelevant items) is usually much larger than the set of positives (relevant items).
  - Discuss: would the ROC curve be affected by class imbalance?
  - Discuss: how would class imbalance affect the effectiveness of ROC?





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The Precision-Recall Curve

- Problems of the ROC curve:
  - Require well-defined “true positives” and “true negatives”;
  - However, in some problems, “negative” may not be well-defined.
  - E.g., **detecting objects** in images: if the detector works by classifying patches, then the # of patches examined — the # of true negatives — is a param of the algorithm, instead of the problem definition.
  - Can we evaluate the classification computed just from positives?

- Precision and the precision-recall curve:
  - Precision: replace the FPR with a quantity that is computed just from positives.

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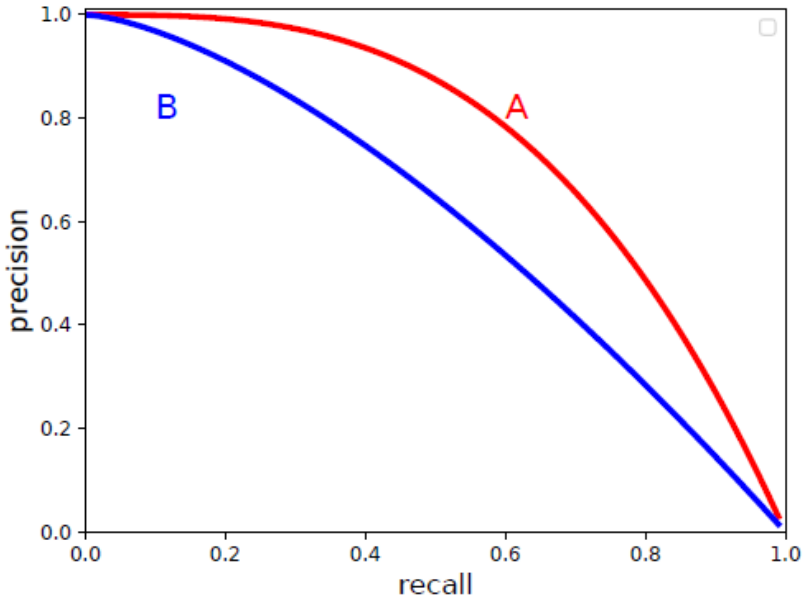
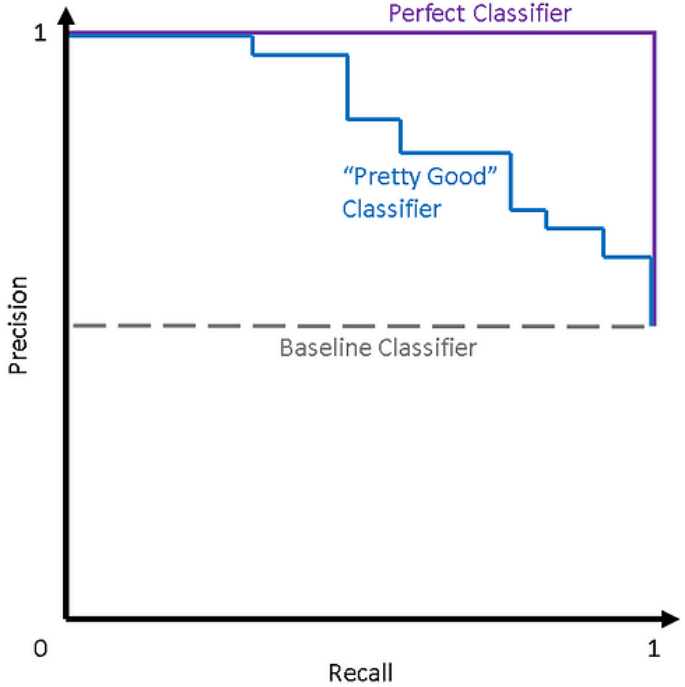
The Precision-Recall (PR) Curve

- Precision: fraction of our detections are actually positive:
- $\mathcal{P}(\tau) \triangleq p(y = 1 | \hat{y} = 1, \tau) = TP_\tau / (TP_\tau + FP_\tau)$
- Recall (TPR): fraction of the positives we actually detected:
- $\mathcal{R}(\tau) \triangleq p(\hat{y} = 1 | y = 1, \tau) = TP_\tau / (TP_\tau + FN_\tau)$





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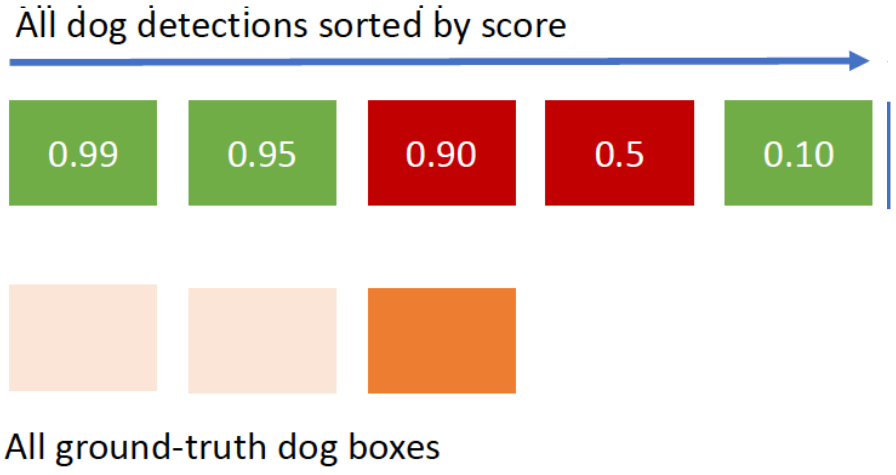
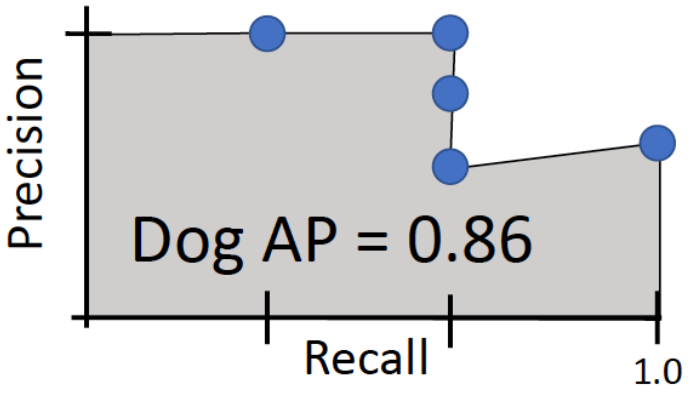
An Example of PR Curve in Object Detection

- Run object detector on all test images (with NMS).
- For each **category**, compute Average Precision (AP) = area under Precision vs Recall Curve.
  - For each detection (highest score to lowest score):
  - If it matches some GT box with IoU > 0.5, mark it as positive and eliminate the GT.
  - Otherwise mark it as negative.
  - Plot a point on PR Curve.
  - Average Precision (AP) = area under PR curve
- Mean Average Precision (mAP) = average of AP for each category.
- For "COCO mAP": Compute mAP @ thresh for each IoU threshold (0.5, 0.55, 0.6, ..., 0.95) and take average.





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mAP@0.5 = 0.77

mAP@0.55 = 0.71

mAP@0.60 = 0.65

...

mAP@0.95 = 0.2

COCO mAP = 0.4

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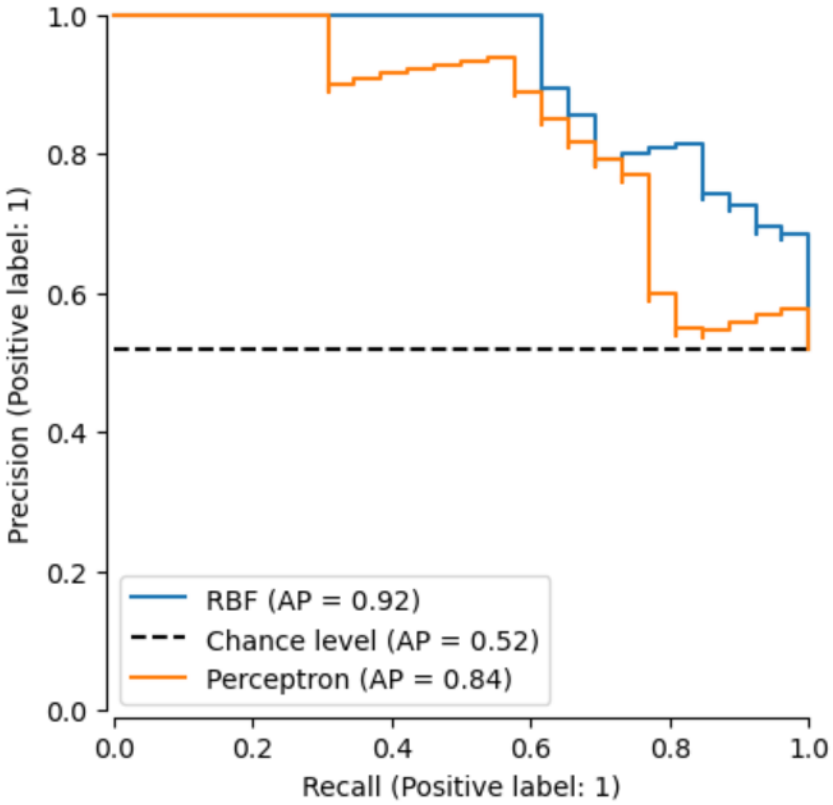
Plotting the PR Curve: Example

- PR curve – Plot the precision vs recall as an implicit function of τ, i.e., plot precision vs recall with varying τ.
- How do we compare different PR curves?



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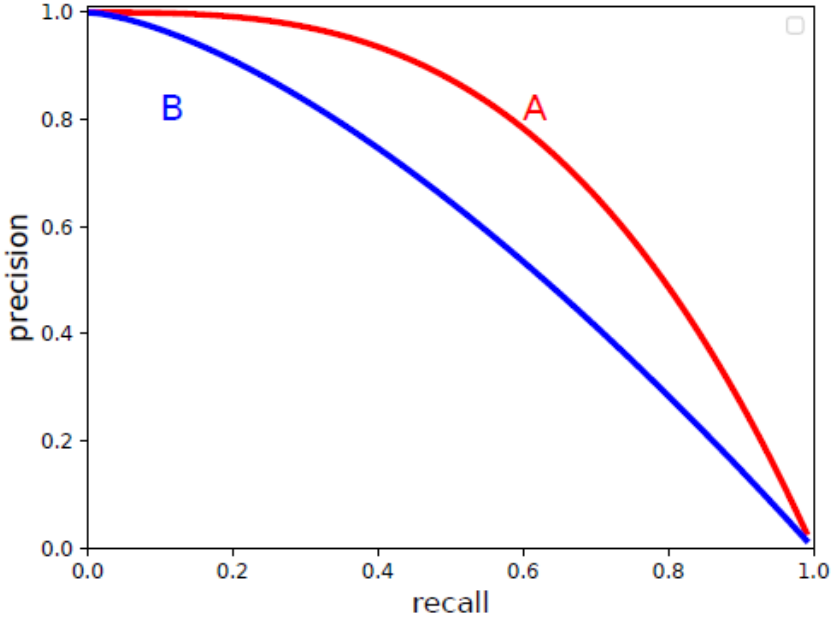
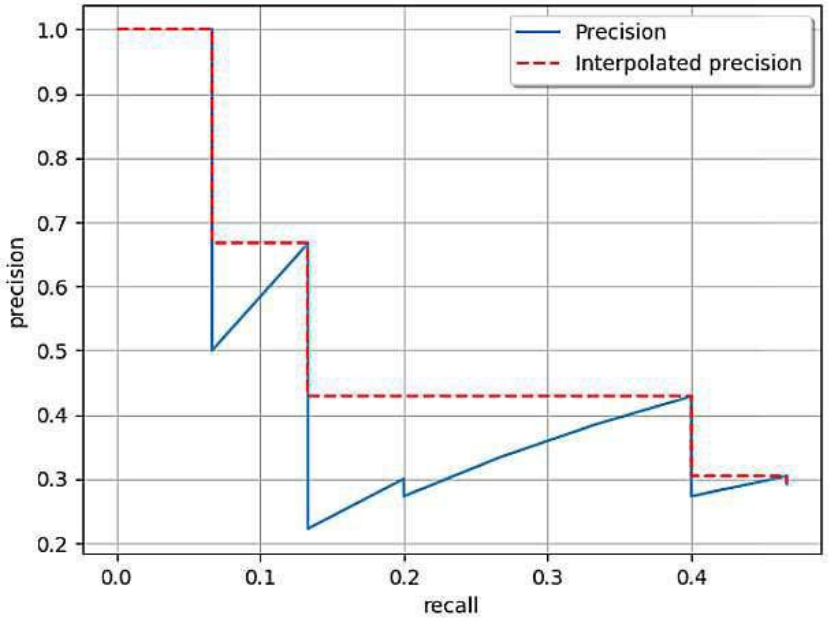
The PR Curve

- The statistics to summarize the PR curve:
  - Precision at K score: the precision for a fixed recall level;
  - AUC: the area under the PR curve;
- **Interpolated precision**:
  - Possible precision **does not drop monotonically** with recall. E.g., a classifier has 90% precision at 10% recall, and 96% precision at 20% recall.
  - Measure the **maximum** precision we can achieve with **at least a recall of 10%** (which would be 96%).
- **Average precision**: avg of interpolated precisions
  - Equal to the area under the interpolated PR curve.
  - The **mean average precision** (mAP): the mean of the AP over a set of different PR curves.





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The PR Curve and F-scores

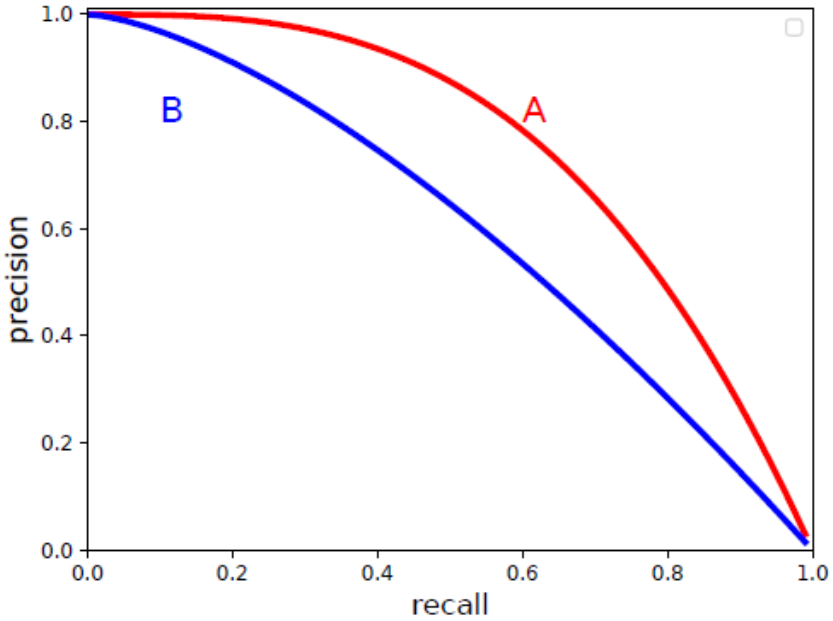
- For a given point on the PR curve, can combine the precision value $\mathcal{P}$ and the recall value $\mathcal{R}$ into a single statistics:

$$\frac{1}{F_\beta} = \frac{1}{1+\beta^2}\frac{1}{\mathcal{P}} + \frac{\beta^2}{1+\beta^2}\frac{1}{\mathcal{R}}$$

- If we set $\beta = 1$, then we get the harmonic mean of the precision and recall, known as the F_1 score.

$$\frac{1}{F_1} = \frac{1}{2}\left(\frac{1}{\mathcal{P}} + \frac{1}{R}\right)$$

- Questions:
  - Why do we use the "harmonic mean" instead of the "arithmetic mean" $(\mathcal{P}+\mathcal{R})/2$?
  - Would the PR curve be affected by class imbalance?



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Probabilistic Prediction Problem

- In previous slides: assumed the set of possible actions was to pick a single class label (or possibly the "reject" action).

- More often: the set of possible **actions** is to pick a **probability distribution** over some value of interest:
  - Perform probabilistic prediction, rather than predicting a specific value.
  - For Classification: states of nature → class labels: $\mathcal{H} = \mathcal{Y} = \{1, \dots, C\}$; actions → class labels: $\mathcal{A} = \mathcal{Y}$.
  - For Probabilistic prediction: states of nature → $h = p(Y|x)$; action → $a = q(Y|x)$.
  - What we want: to pick q to minimize$\mathbb{E}[\ell(h, a)] = \mathbb{E}[\ell(p, q)]$ for a given x.
  - The key: how to define the loss function $\ell(p, q)$?

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Bayesian Learning Fundamentals 2:

- ✓ KL-Divergence
- ✓ Entropy and Mutual Information

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KL Divergence (KL-Div)

- Loss function is measured between two “distributions”.
- Kullback Leibler divergence (**KL divergence**) is a measure of how one probability distribution is different from a second:

$$\mathbb{D}_{KL}(p \parallel q) \triangleq \sum_{y^* \in \mathcal{Y}} p(y^*) \log \frac{p(y^*)}{q(\hat{y})}$$

- Here p is the true distribution and q is the predicted distribution.
- KL divergence can be further expanded as:
- $\mathbb{D}_{KL}(p \parallel q) = \sum p(y^*) \log p(y^*) - \sum p(y^*) \log q(\hat{y}) = -\mathbb{H}(p) + \mathbb{H}_{ce}(p, q)$. $\mathbb{H}(p)$: entropy of p; $\mathbb{H}_{ce}$: cross entropy between p and q.

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Entropy

- The key component of KL Divergence: entropy.
  - The **entropy** of a **probability distribution** can be interpreted as a measure of uncertainty, or lack of predictability, associated with a random variable drawn from a given distribution.
  - Entropy: define the **information content** of a data source.
  - Suppose we observe a sequence of symbols $X_n \sim p$ generated from distribution p. What does it mean by p having high / low entropy?
  - If p has high entropy, hard to predict the value of each observation X_n. Hence dataset $\mathcal{D} = (X_1, \dots, X_n)$ has high information content.
  - If p is a degenerate distribution with 0 entropy, then every X_n will be the same, so $\mathcal{D}$ does not contain much information.

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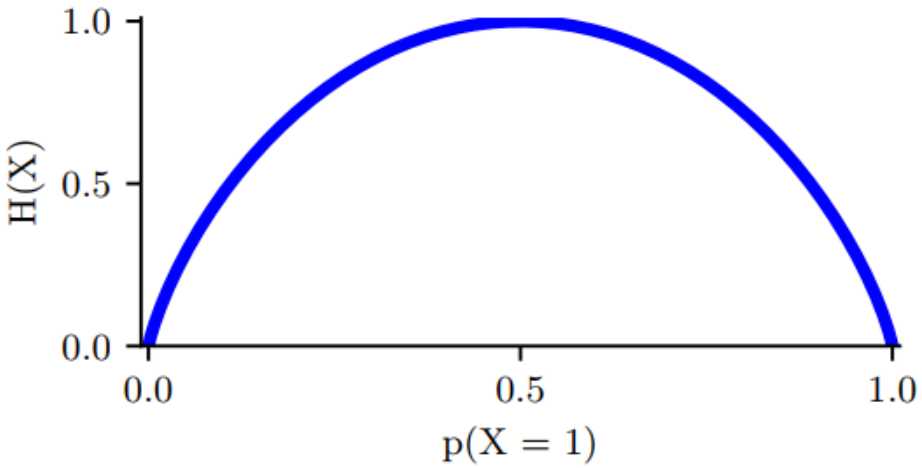
Entropy for Discrete Random Variables

- More specifically, formulate the entropy of a discrete RV X with distribution p over K states (classes) as:

$$\mathbb{H}(X) \triangleq -\sum_{k}^{K} p(X=k)\log_2 p(X=k) = -\mathbb{E}_X[\log p(X)]$$

  - Note that $\mathbb{H}(X)$ is equivalent to the notation of $\mathbb{H}(p)$.
  - Question: when will the distribution obtain the Max / Min Entropy?



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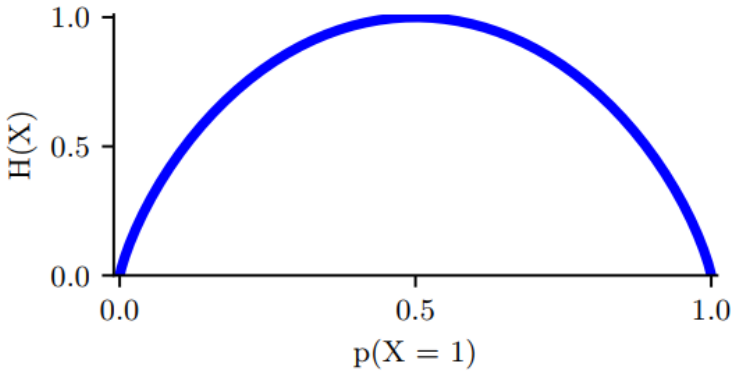
Entropy for Discrete Random Variables

$$\mathbb{H}(X) \triangleq -\sum p(X=k)\log_2 p(X=k) = -\mathbb{E}_X[\log p(X)]$$

- Discrete distribution with Maximum entropy is uniform distribution.
- For a K-ary RV, the entropy is max-ed if $p(x=k) = 1/K$; $H(X) = \log_2 K$.
- Distribution with min entropy is any **delta-function** that puts all its mass on one state

- Special case: binary RVs $X \in \{0,1\}$, $p(X=1) = \theta$ and $p(X=0) = 1-\theta$:
  - $\mathbb{H}(X) = -[p(X=1)\log_2 p(X=1) + p(X=0)\log_2 p(X=0)]$
  - $\mathbb{H}(X) = -[\theta \log_2 \theta + (1-\theta)\log_2(1-\theta)]$



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Mutual Information

- KL divergence: measure how **dissimilar** two distributions were.
- Then can we measure how **dependent** two RVs are?
  - Turn the question of measuring the **dependence** of two RVs into a question about the **similarity** of their distributions.
- The notion of Mutual Information (MI) btw two RVs, defined as:

$$\mathbb{I}(X;Y) \triangleq \mathbb{D}_{KL}\big(p(x,y) \parallel p(x)p(y)\big) = \sum_{y}\sum_{x} p(x,y)\log\frac{p(x,y)}{p(x)p(y)}$$

  - Note: the notation is defined as $\mathbb{I}(X;Y)$ as X and/or Y can represent a set of variable – $\mathbb{I}(X;Y,Z)$ is the MI between X and (Y,Z).
  - MI is always non-negative: $\mathbb{I}(X;Y) \geq 0$. Reach the bound of 0 iff $p(x,y) = p(x)p(y)$.

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Mutual Information: Interpretation

$$\mathbb{I}(X;Y) \triangleq \mathbb{D}_{KL}\big(p(x,y) \parallel p(x)p(y)\big) = \sum_{y}\sum_{x} p(x,y)\log\frac{p(x,y)}{p(x)p(y)}$$

- MI is a KL-Div between the joint and factored marginal distributions:
- MI measures the **information gain** if we update from a model that treats the two variables as independent $p(x)p(y)$ to one that models their true joint density $p(x,y)$.
- To gain further, re-express MI in terms of joint and conditional entropies:
  $\mathbb{I}(X;Y) = \mathbb{H}(X) - \mathbb{H}(X|Y) = \mathbb{H}(Y) - \mathbb{H}(Y|X), \mathbb{H}(Y|X) = \mathbb{H}(X,Y) - H(X)$
- MI between X and Y is thus expressed as the reduction in uncertainty about X after observing Y, or, by symmetry, the reduction in uncertainty about Y after observing X.

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Mutual Information: Interpretation

$$\mathbb{I}(X;Y) \triangleq \mathbb{D}_{KL}\big(p(x,y) \parallel p(x)p(y)\big) = \sum_{y}\sum_{x} p(x,y)\log\frac{p(x,y)}{p(x)p(y)}$$

$$\mathbb{I}(X;Y) = \mathbb{H}(X) - \mathbb{H}(X|Y) = \mathbb{H}(Y) - \mathbb{H}(Y|X), \mathbb{H}(Y|X) = \mathbb{H}(X,Y) - H(X)$$

- An alternative proof that conditioning, on average, reduces entropy!

$$\mathbb{I}(X;Y) = \mathbb{H}(X) - \mathbb{H}(X|Y) \geq 0 \rightarrow \mathbb{H}(X|Y) \leq \mathbb{H}(X)$$

- One can also obtain the following equivalent results of MI:

$$\mathbb{I}(X;Y) = \mathbb{H}(X,Y) - \mathbb{H}(X|Y) - \mathbb{H}(Y|X)$$

$$\mathbb{I}(X;Y) = \mathbb{H}(X) + \mathbb{H}(Y) - \mathbb{H}(X,Y)$$

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Mutual Information: An Example

- Suppose two RVs X and Y are related to the input integer $n \in \{1, \dots, 8\}$.
  - Define $X(n) = 1$ if n is even and $Y(n) = 1$ if n is a prime number. Thus:

n	1	2	3	4	5	6	7	8
X	0	1	0	1	0	1	0	1
Y	0	1	1	0	1	0	1	0

$p(X,Y)$	$Y=0$	$Y=1$
$X=0$	$\frac{1}{8}$	$\frac{3}{8}$
$X=1$	$\frac{3}{8}$	$\frac{1}{8}$

	Y=0	Y=1
X=0	$\frac{1}{4}$	$\frac{3}{4}$
X=1	$\frac{3}{4}$	$\frac{1}{4}$

  - $\mathbb{H}(X,Y) = -\left[\frac{1}{8}\log\frac{1}{8} + \frac{3}{8}\log\frac{3}{8} + \frac{3}{8}\log\frac{3}{8} + \frac{1}{8}\log\frac{1}{8}\right] = 1.81\ bits$
  - $\mathbb{H}(X) = \mathbb{H}(Y) = 1$
  - $\mathbb{H}(Y|X) \triangleq -\sum p(X,Y)\log p(Y|X) = -\left[\frac{1}{8}\log\frac{1}{4} + \frac{3}{8}\log\frac{3}{4} + \frac{3}{8}\log\frac{3}{4} + \frac{1}{8}\log\frac{1}{4}\right] = 0.81$
  - Verify that $\mathbb{H}(Y|X) = \mathbb{H}(X,Y) - H(X) = 0.81$
  - $\mathbb{I}(X;Y) = \mathbb{H}(X) - \mathbb{H}(X|Y) = 1 - 0.81 = 0.19\ bits$

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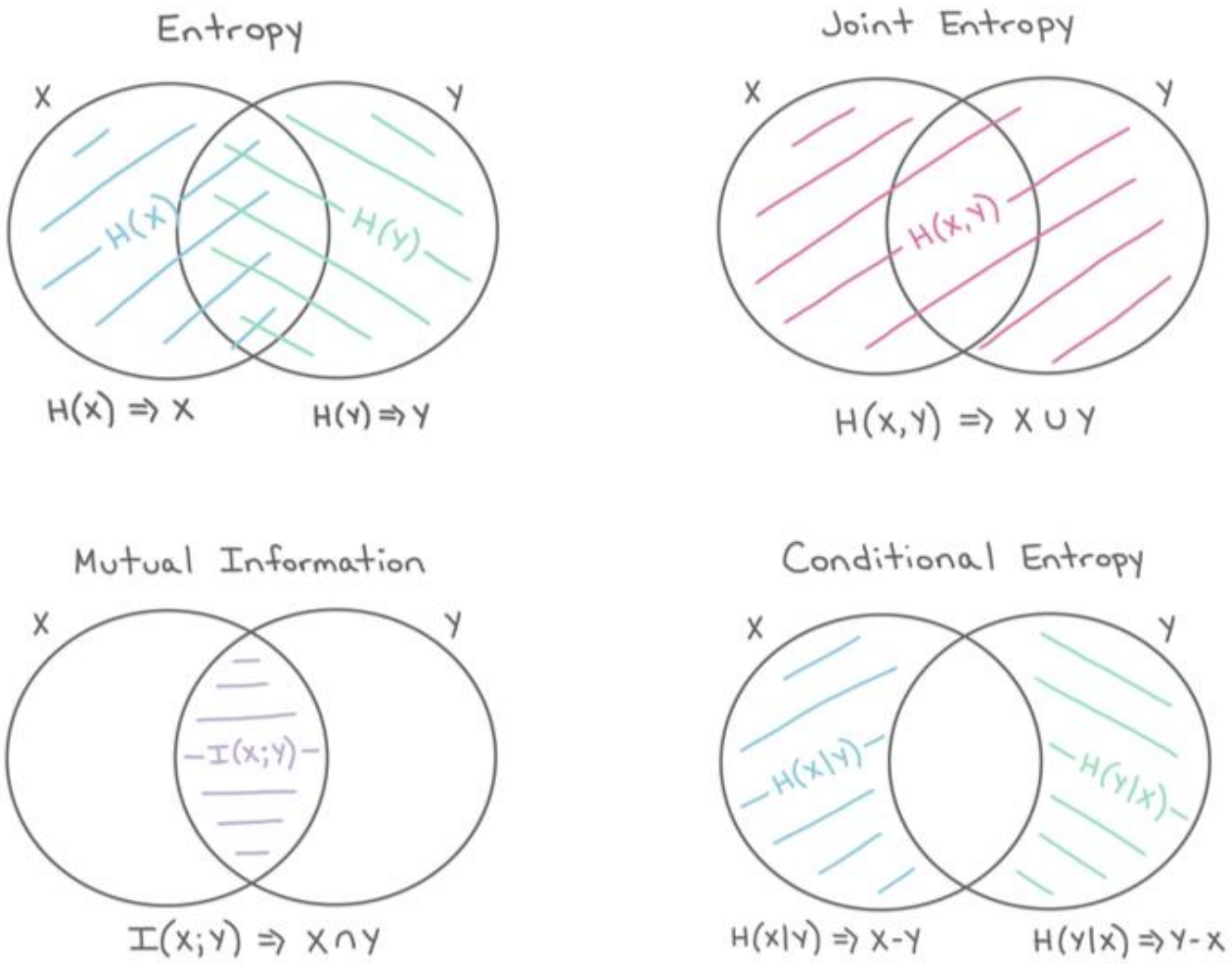
Entropy and Mutual Information



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Bayesian Learning Fundamentals 3:

- ✓ The Frequentist Decision
- ✓ Application in Supervised Learning

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Recall: Frequentist vs. Bayesian

- Frequentist interpretation: probabilities represent long run frequencies of events that can happen multiple times.
- E.g., Flipping the coin many times, we expect it to land heads about half the time.
- Bayesian interpretation: quantify **uncertainty** or ignorance about something
- E.g., The coin is equally likely to land heads/tails on next toss.

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Frequentist Decision Theory

- In the **Frequentist decision theory**, treat the **unknown** state of nature (denoted by $\boldsymbol{\theta}$ instead of h) as a **fixed yet unknown quantity**, and treat the data x as random.
  - Instead of conditioning over x, we **average** over it to compute the loss expect to incur if apply our estimator to **different datasets**.

- Define the frequentist **risk** of an estimator δ given an unknown state of nature $\boldsymbol{\theta}$ as the expected loss when applying δ to data x, where the expectation is over the data, sampled from $p(x|\boldsymbol{\theta})$:

$$R(\boldsymbol{\theta}, \delta) \triangleq \mathbb{E}_{p(x|\boldsymbol{\theta})}\left[\ell\left(\boldsymbol{\theta}, \delta(x)\right)\right]$$

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The Bayes Risk

$$R(\boldsymbol{\theta}, \delta) \triangleq \mathbb{E}_{p(x|\boldsymbol{\theta})}\left[\ell\left(\theta, \delta(x)\right)\right]$$

- In general, the true state of nature $\boldsymbol{\theta}$ that generates the data x is unknown, so cannot compute the risk directly. Then how?
- One solution: to assume a prior π_0 for θ then average it out – the **Bayes Risk** (integrated risk):

$$R_{\pi_0}(\delta) \triangleq \mathbb{E}_{\pi_{0(\boldsymbol{\theta})}}[R(\boldsymbol{\theta}, \delta)] = \int d\boldsymbol{\theta} dx \pi_0(\boldsymbol{\theta}) p(x|\boldsymbol{\theta}) \ell\left(\boldsymbol{\theta}, \delta(x)\right)$$

- Decision rule that minimize the Bayes risk: **Bayes estimator**.

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The Bayes Estimator

- Recall: in the Bayesian decision theory, we define the **optimal policy**, also the **Bayes estimator** or **Bayes decision rule** as:

$$\pi^*(x) = \arg\min_{a \in \mathcal{A}} \rho(a|x) = \arg\min_{a \in \mathcal{A}} \mathbb{E}_{p(h|x)}[\ell(h, a)]$$

- In the Frequentist decision theory, the optimal estimator is:

$$\delta(x) = \arg\min_{a} \int d\theta \pi_0(\theta) p(x|\theta) \ell(\theta, a) \equiv \arg\min_{a} \int d\theta p(\theta|x) \ell(\theta, a)$$

- The two forms are actually equivalent! What does this mean?
  - Bayesian approach provides a good way of achieving the frequentist goal – picking the optimal action case-by-case is optimal on average!

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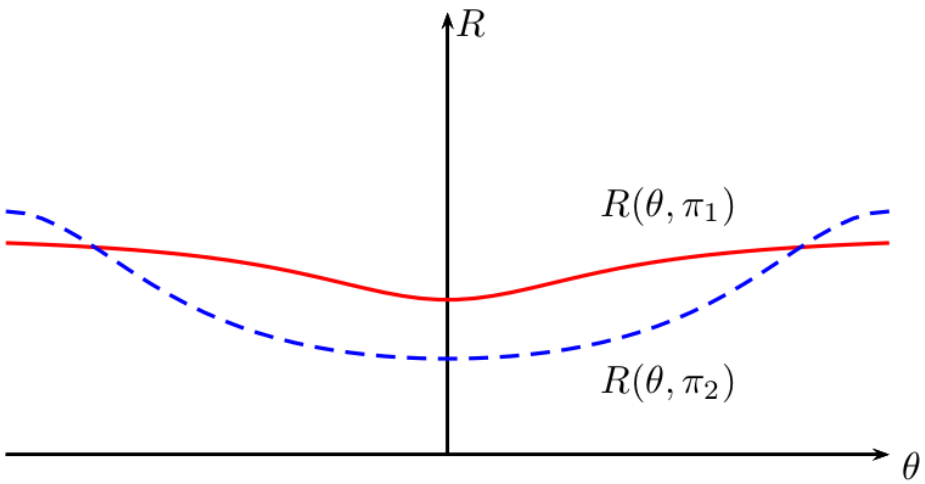
The Maximum Risk

- Use of a prior might seem undesirable in certain cases. We can therefore define the **maximum risk** as:

$$R_{\max}(\delta) \triangleq \sup_{\boldsymbol{\theta}} R(\boldsymbol{\theta}, \delta)$$

- Decision rule that min the max risk: minimax estimator δ_{MM}.

- One example on the right:
  - δ_1 has lower worst-case risk than δ_2.
  - δ_1 is the minimax estimator (even if δ_2 has lower risk for most values of $\boldsymbol{\theta}$)/
  - The minimax estimators are overly conservative!



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ERM and Frequentist Decision Theory

- We consider how to apply the frequentist decision theory in the context of **supervised learning**.
- The risk: $R(\boldsymbol{\theta}, \delta) \triangleq \mathbb{E}_{p(x|\boldsymbol{\theta})}\left[\ell\left(\boldsymbol{\theta}, \delta(x)\right)\right]$.
  - Unknown "state of nature" can correspond to the unknown **parameters** of some model $\boldsymbol{\theta}^*$.
  - Meanwhile, the discussion w.r.t. data within a dataset $x \in \mathcal{D}$, the risk can be rewritten:

$$R(\delta, \boldsymbol{\theta}^*) = \mathbb{E}_{p(\mathcal{D}|\theta^*)}\left[\ell\left(\boldsymbol{\theta}^*, \delta(\mathcal{D})\right)\right]$$

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ERM and Frequentist Decision Theory

- In supervised learning,
  - Different unknown state of nature (output y) for each different input x,
  - Estimator δ is a prediction function $\hat{y} = f(x)$,
  - The state of nature (as a whole) is the true distribution $p^*(x, y^*)$,
  - Thus, the risk of an estimator is (in terms of f and p^*) is:
  - $R(f, p^*) = R(f) \triangleq \mathbb{E}_{p^*(x)p^*(y|x)}[\ell(y^*, f(x))]$
  - Defined as the **population risk**, as the expectations are taken w.r.t. the true joint distribution $p^*(x, y^*)$.
- Can we compute the **population risk**? If not how?

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ERM and Frequentist Decision Theory

- In most cases, quite impossible to obtain the unknown true distribution p^*.
  - Instead, can approximate it with the empirical distribution of N samples:
  - $p_{\mathcal{D}}(x, y|\mathcal{D}) \triangleq \frac{1}{|\mathcal{D}|}\sum_{(x_n,y_n)\in\mathcal{D}} \delta(x - x_n)\delta(y - y_n)$
  - Here the empirical distribution is exactly the training distribution $p_{\mathcal{D}}(x, y|\mathcal{D}) = p_{tr}(x, y)$. We thus obtain the empirical risk (which is a random variable) as:

$$R(f, \mathcal{D}) \triangleq \mathbb{E}_{p_{\mathcal{D}}(x, y|\mathcal{D})}[\ell(y, f(x)] = \frac{1}{N}\sum_{n=1}^{N} \ell\big(y_n, f(x_n)\big)$$

  - A natural way to choose the predictor is to choose the predictor with minimum risk:
  - $\hat{f}_{ERM} = \arg\min_f R(f, \mathcal{D}) = \arg\min_f \frac{1}{N}\sum_n^N \ell(y_n, f(x_n)) = \arg\min_\theta \mathcal{L}(\theta)$
  - $\mathcal{L}(\boldsymbol{\theta}) = \sum_{n=1}^N \ell(y_n, \boldsymbol{\theta}; x_n)$ (as defined in Lecture 1)
- The process of choosing predictor $\hat{f}_{ERM}$ is **empirical risk minimization (ERM)**.

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Performance of Functions Fitted with ERM

- What is the theoretical performance of functions that are fitted with the ERM principle?
  - Let $f^{**} = \arg\min_{f} R(f)$ be the function that achieves the min possible population risk, where we optimize over ALL possible functions.
  - !! **ALL possible functions**?
  - Define $f^{*} = \arg\min_{f \in \mathcal{H}} R(f)$ the best function in hypothesis space $\mathcal{H}$.
  - We still cannot compute what f^{*} is – the population risk R is unknown!
  - We can only compute by further approx. – minimizing the empirical risk:

$$f_N^{*} = \arg\min_{f \in \mathcal{H}} R\left(f, \mathcal{D}\right) = \arg\min_{f \in \mathcal{H}} \mathbb{E}_{p_{tr}}\left[\ell\left(y, f(x)\right)\right]$$

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Performance of Functions Fitted with ERM

- The risk of our chosen predictor f_N^* compared to the best possible predictor f^{**} can be decomposed into two terms:
- $\mathbb{E}_{p^*}[R(f_N^*) - R(f^{**})] = \underbrace{R(f^*) - R(f^{**})}_{\varepsilon_{app}(\mathcal{H})} + \underbrace{\mathbb{E}_p[R(f_N^*) - R(f^*)]}_{\varepsilon_{est}(\mathcal{H}, N)}$
  - $\varepsilon_{app}(\mathcal{H})$ is the approximate error: measures how closely $\mathcal{H}$ can model the true optimal function f^{**}.
  - $\varepsilon_{est}(\mathcal{H}, N)$ is the estimation error or generalization error: measures the difference in estimated risk due to having a finite training set.
  - Approx. $\varepsilon_{est}(\mathcal{H}, N)$ by the difference btw training set and test set error: $\varepsilon_{est}(\mathcal{H}, N) \approx \mathbb{E}_{p_{tr}}[\ell(y, f_N^*(x))] - \mathbb{E}_{p_{te}}[\ell(y, f_N^*(x))]$ – Generalization Gap.

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Regularization and Regularized Risk

- To avoid the chance of overfitting, it is common to add a complexity penalty to the objective function – regularization.
  - With regularization, we obtain the **regularized** empirical risk:
  - $R_\lambda(f, \mathcal{D}) = R(f, D) + \lambda C(f)$
  - $C(f)$ measures the complexity of the prediction function f and $\lambda \geq 0$ is the hyperparameter that controls the strength of complexity penalty.
  - In practice, the regularization is often applied to the model parameters directly, thus the regularized ER is rewritten: $R_\lambda(\theta, \mathcal{D}) = R(\theta, D) + \lambda C(\theta)$.
  - If the loss function is a log loss term, while the regularizer is a negative log prior, $R_\lambda(\theta, \mathcal{D}) = -\frac{1}{N}\sum_{n=1}^{N} \log p(y_n | x_n, \theta) - \lambda \log p(\theta)$. Minimizing this is exactly equivalent to the MAP estimation.

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Regularization and Regularized Risk

- To avoid the chance of overfitting, it is common to add a complexity penalty to the objective function – regularization.

- Recall when we first introduce the MAP estimation:
- The action that minimizes the expected loss is to choose the most probable action:

$$\pi(x) = \arg\max_{\hat{y} \in \mathcal{Y}} p(\hat{y}|x)$$

- This corresponds to the **mode** of the posterior distribution, also known as the **maximum a posteriori** or **MAP** estimate.

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Occam's Razor

- **Occam's razor** is the problem-solving principle that recommends searching for explanations constructed with the smallest possible set of elements.
  - Also known as the **principle of parsimony** or the **law of parsimony.**
  - Consider 2 models, a simple one, m_1, and a more complex one, m_2. Suppose both can explain the data by suitably optimizing their parameters, i.e., $p(\mathcal{D}|\widehat{\boldsymbol{\theta}}_1, m_1)$ and $p(\mathcal{D}|\widehat{\boldsymbol{\theta}}_2, m_2)$ are both large.
- Intuitively should prefer m_1, as it is **simpler** and as good as m_2.
- In regularization, it corresponds to the complexity penalty, where more complex models would result in a larger regularization term.

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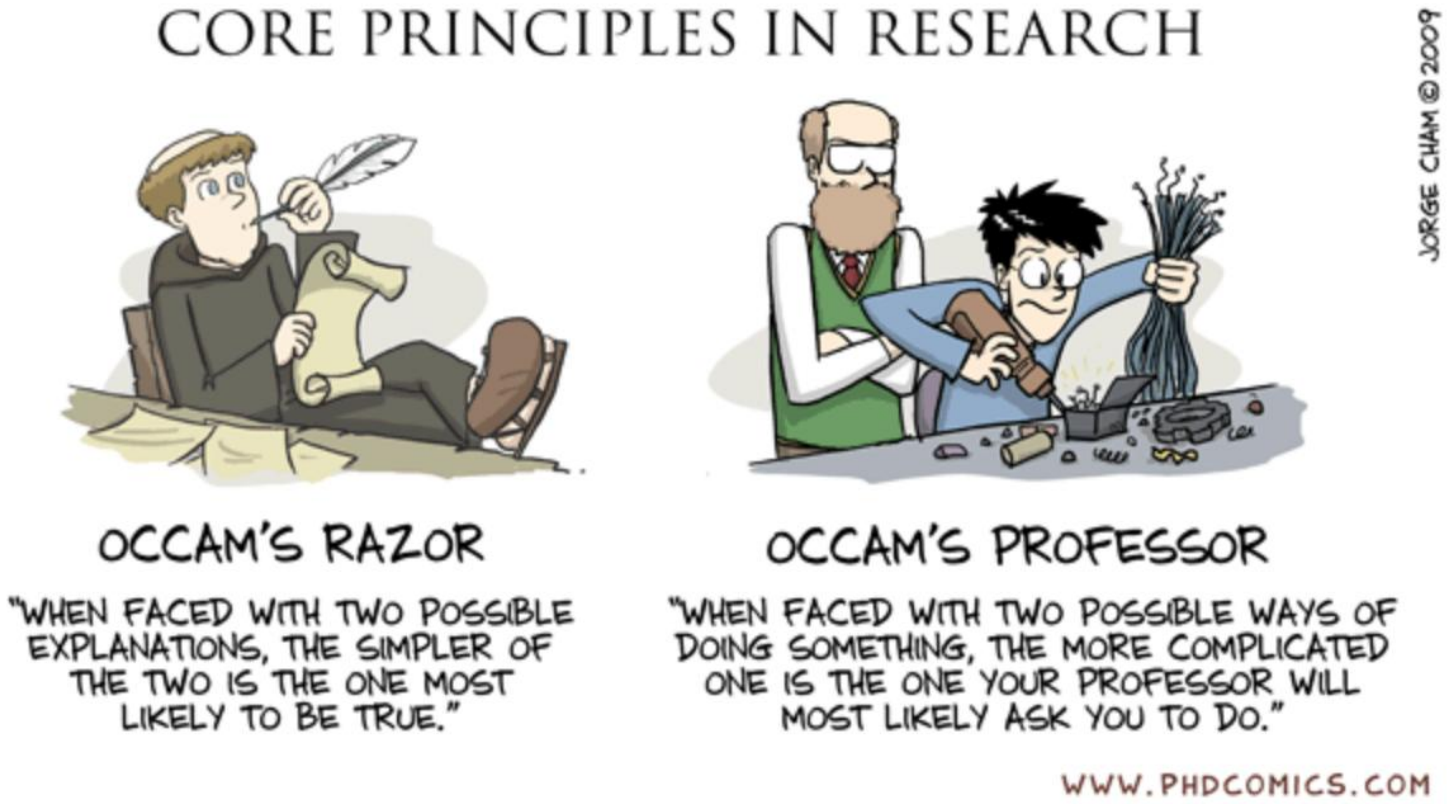
Occam's Razor

Piled Higher and Deeper ***by Jorge Cham*** **www.phdcomics.com**



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Population Risk Estimation with Validation

- A simple way to estimate the population risk for a supervised learning setup: cross-validation.
  - Partition the dataset into two, one used for training models (training set), another, the **validation** set or **holdout** set, used for assessing the risk.
  - Note that in supervised learning, the real "test data" are **inaccessible**.
- Process: fit the model on the training set, use its performance on the validation set as an approx. to the population risk:
  - Make the dependence of the $R_\lambda(f, \mathcal{D})$ on the dataset more explicit:

$$R_\lambda(\boldsymbol{\theta}, \mathcal{D}) = \frac{1}{|D|} \sum_{(x,y)\in\mathcal{D}} \ell\big(y, f(x; \boldsymbol{\theta})\big) + \lambda C(\boldsymbol{\theta})$$

  - Also define $\widehat{\boldsymbol{\theta}}_\lambda(\mathcal{D}) = \arg\min_\theta R_\lambda(\theta, \mathcal{D})$, $\mathcal{D}_{tr}$ and $\mathcal{D}_{val}$ are partitions.

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Cross-Validation

- For each model, we fit it to the training set to obtain $\widehat{\boldsymbol{\theta}}_\lambda(\mathcal{D}_{tr})$. Then use the **unregularized** empirical risk on the validation set as an estimate of the population risk – validation risk: $R_\lambda^{val} \triangleq R_0\left(\widehat{\boldsymbol{\theta}}_\lambda(\mathcal{D}_{tr}), \mathcal{D}_{val}\right)$.
  - Any shortcomings of this method?
- Consider the training data is very limited – not enough data to train on! Impossible to make reliable estimate of future performance.

- Solution?

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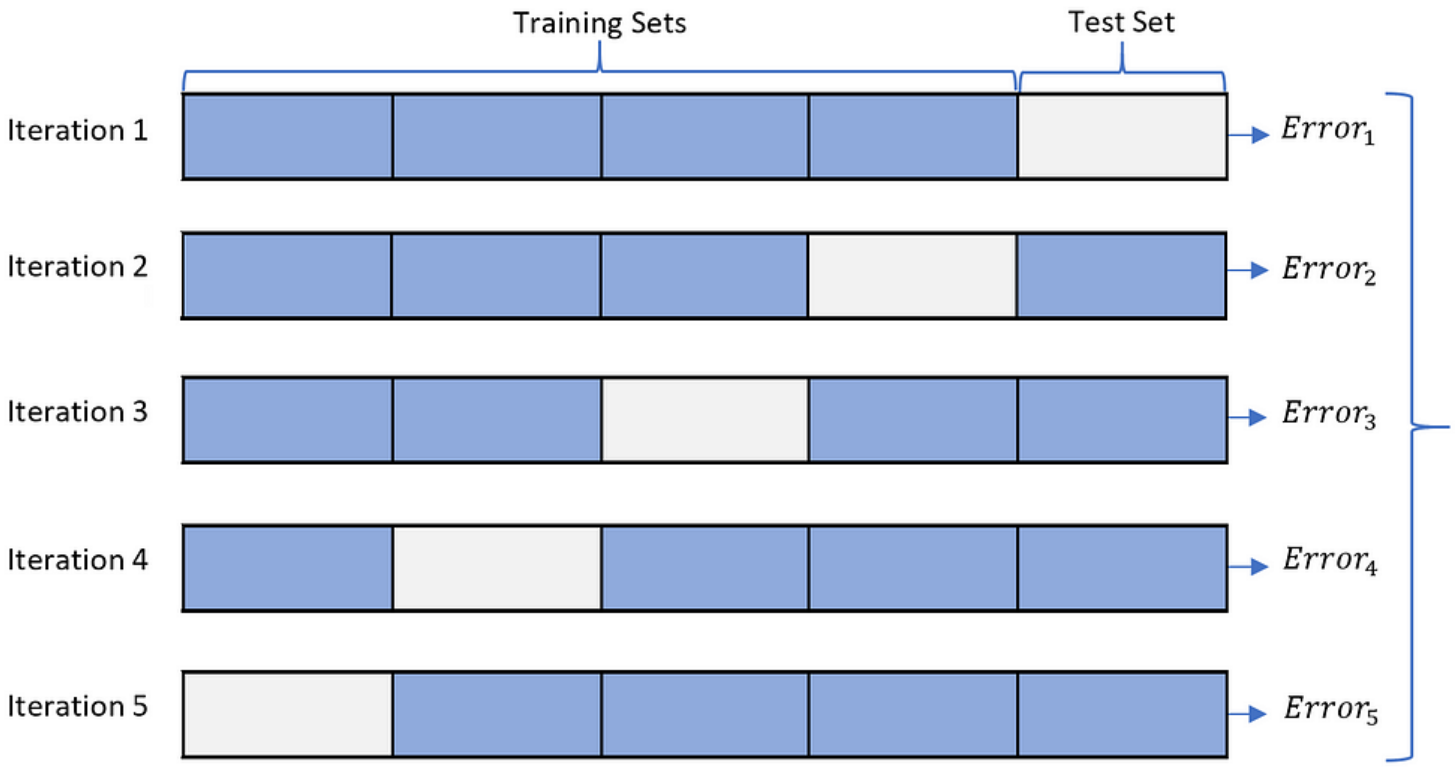
Cross-Validation

- Consider the training data is very limited – not enough data to train on! Impossible to make reliable estimate of future performance.

- Solution?
- Cross-validation!



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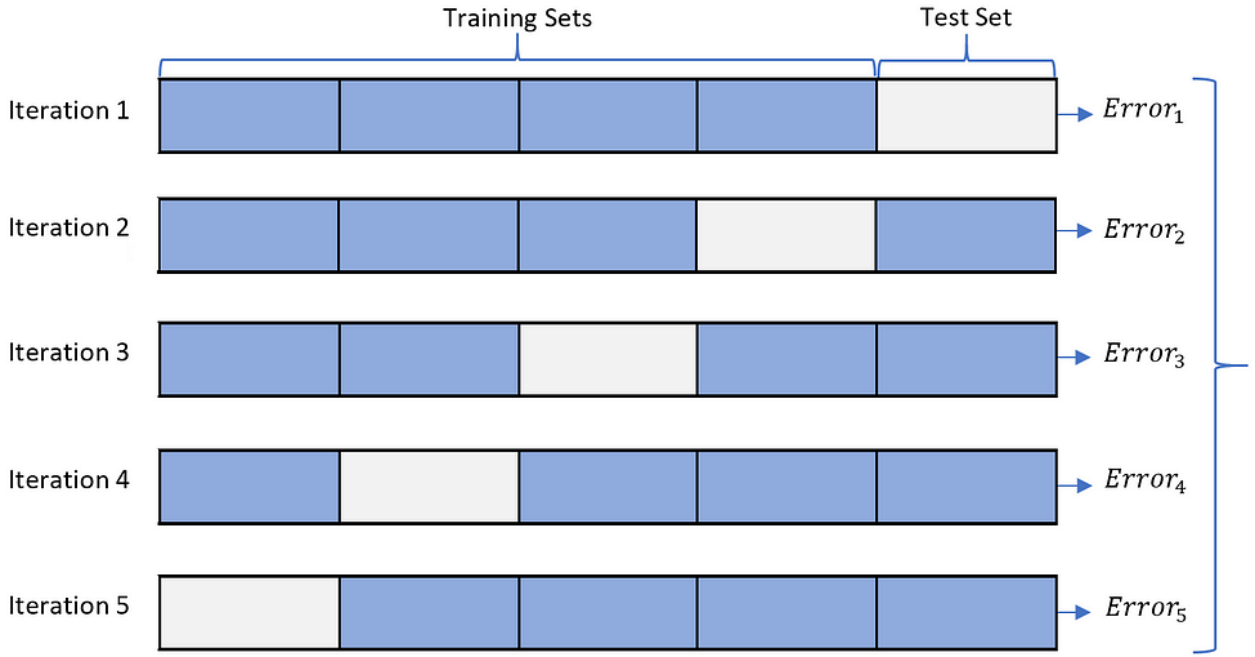
Cross-Validation

- Split the training data into K folds;
- For each fold $k \in \{1, \dots, K\}$, train on all but ONE fold, in a round-robin fashion (in turns, circled around all possible folds).



- Formally, the cross-validation risk is formulated as:
- $R_\lambda^{cv} \triangleq \frac{1}{K}\sum_{k=1}^{K} R_0\left(\widehat{\boldsymbol{\theta}}_\lambda(\mathcal{D}_{-k}), \mathcal{D}_k\right)$
- $\mathcal{D}_k$ is the data in the k-th fold, and $\mathcal{D}_{-k}$ is all the other data. If set $K = N$, the method is known as leave-one-out cross-validation.
- Alternatively, we can estimate an optimal $\hat{\lambda}$ with CV, then re-estimate the model parameters using all data: $\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} R_{\hat{\lambda}}(\boldsymbol{\theta}, \mathcal{D})$.

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Cross-Validation: Example

```python
from sklearn.model_selection import KFold
k = 5
kf = KFold(n_splits=k, shuffle=True, random_state=0)
```

✓ 0.0s

```python
from sklearn.model_selection import cross_val_score
scores_sklearn = cross_val_score(model, X, y, cv=kf, scoring='accuracy')
mean_accuracy_sklearn = np.mean(scores_sklearn)

print(f"Accuracy Score for each fold: {[float(score) for score in scores_sklearn]}")
print(f"Mean accuracy across {k} folds: {mean_accuracy_sklearn}")
```

✓ 0.0s

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Summary

- Fundamentals: Probability and the difference between distributions:
  - Probability and the Bayes' Rule.
  - All sorts of univariate probability distributions.
  - KL-Divergence, Entropy, and Mutual Information.
- Bayesian Learning and Modelling:
  - Modelling with known distribution type: MLE.
  - Empirical Risk Minimization (ERM).
  - Decision making for best prediction: classification problems and MAP.
  - The Bayesian and Frequentist Decision Theories.
  - Regularized risk with cross-validation.

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